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VIRTSI: A novel trust dynamics model enhancing Artificial Intelligence collaboration with human users – Insights from a ChatGPT evaluation study

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A B S T R A C T

The rapid integration of intelligent processes and methods into information systems in the Artificial Intelligence (AI) era has led to a substantial shift towards autonomous software decision-making. This evolution necessitates robust human oversight, especially in critical domains like Healthcare, Education, and Energy. Human trust in AI plays a vital role in influencing decision-making processes of users interacting with AI.

This paper presents VIRTSI (**V**ariability and **I**mpact of **R**eciprocal **T**rust **S**tates towards **I**ntelligent systems), a novel rigorous computational model for human-AI Interaction. VIRTSI simulates human trust states, spanning from overtrust to distrust, through user modelling. It comprises: 1. A trust dynamics representational model based on Deterministic Finite State Automata (DFAs), illustrating transitions among cognitive trust states in response to AI-generated replies. 2. A trust evaluation model based on Confusion Matrices, originating from machine learning and Accuracy Metrics, providing a quantitative framework for analysing human trust dynamics. As a result, this is the first time that trust dynamics have been thoroughly traced in a representational model and a method has been developed to assess the impact of possibly harmful states like overtrust and distrust.

An empirical study on the recently launched Large Language Model of generative AI, ChatGPT (version 3.5), provides a radical underexplored AI-generated platform for evaluating the human-AI interaction through VIRTSI. The study involved 1200 interactions of real users as well as AI experts together with experts in two very different domains of evaluation, namely software engineering and poetry. This study traces trust dynamics and the emerging human-AI interaction, in concrete examples of real user synergies with generative AI. The research reveals the vital role of maintaining normal trust states for optimal human-AI interaction and that both AI and human users need further steps towards this goal. The real-world implications of this research can guide the creation and evaluation of user interfaces with AI and the incorporation of functionalities in the development of generative AI chatbots in terms of trust by providing a new rigorous DFA representational method of trust dynamics and a corresponding new perspective of confusion matrix evaluation method of the dynamics' impact in the efficiency of human-AI dialogues.

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1. Introduction

The rapid growth of Artificial Intelligence (AI) is transforming the role of computers, shifting from deterministic computing to dynamic and interactive AI-driven decision-making. Unlike traditional computing, AI introduces autonomy, making decisions based on predictions and learning from data. This shift is evident in critical decision-making fields in current intelligent information systems and applications like in Healthcare, Education, and Energy Sustainability [6,42]. However, the integration of efficient human-AI interaction poses challenges, such as the potential consequences of human errors in trusting or rejecting AI-generated decisions. Trust in AI-empowered software becomes a vital question, as cognitive states of trust influence the accuracy of human decisions. The complexity arises from inadequate human preparation for this paradigm shift, leading to preconceptions and problematic attitudes like overtrust, mistrust, and distrust in AI. Addressing these challenges is essential for effective human-AI interaction in critical decision-making processes.

While there is considerable research focused on further enhancing algorithms for autonomous reasoning in AI's technical aspects, there exists an extensive under-researched gap in the common understanding of the collaboration between humans and AI, particularly in the context of autonomous decision-making. This stresses the emerging need for more exploration and study in human-AI interaction to illuminate how these entities can effectively collaborate in decision-making processes. This exploration may require user modelling techniques that simulate cognitive aspects and reasoning about trust in human-AI interaction.

User modelling involves creating representations of individual users' beliefs and cognitive aspects by integrating inputs from standalone software and web-based platforms across multiple applications, as discussed in [48]. Extensive exploration of cognitive theories and users' aspects has been undertaken to simulate human reasoning. This exploration enables the generation of personalised advice for human users, addressing areas such as fuzzy reasoning for cognitive levels of users' knowledge, cognitive theories for intelligent interaction or for considering human emotions in affective computing [1,2,7,11]. In the context of user modelling there have been recent research approaches concerning cognitive and affective trust in human–computer interaction. However, in the field of human-AI interaction, it has been highlighted by researchers that fully autonomous system behaviour fails to establish an adequate human–computer trust relationship, in contrast to conservative strategies [25]. In yet more recent advances, it has been noted that the design of adequate proactivity for AI-based systems to support humans is still an open question and a challenging topic [24].

As is evident from the above, cognitive trust user modelling has not been adequately explored in the context of human-AI interaction as is our aim in the present paper. To address this critical research gap, this paper introduces VIRTSI (**V**ariability and **I**mpact of **R**eciprocal **T**rust **S**tates towards **I**ntelligent systems), a novel computational framework designed to simulate and model the dynamics of human cognitive states of trust in human-AI Interaction. The term "virtual" emphasises the computational nature of the model, where human cognitive processes, particularly those related to trust, are simulated. The model encompasses diverse trust states, ranging from normal to problematic states such as overtrust, mistrust, and distrust, all rooted in the cognitive processes of human-system interactions.

Furthermore, VIRTSI introduces an innovative approach to constructing confusion matrices within its framework. Confusion matrices were originally employed in machine learning but, in the context of VIRTSI, they are uniquely applied within the virtual human finite state machine to assess correctness of human decisions regarding the acceptance or rejection of AI-generated responses. This equips VIRTSI with a quantitative model for the analysis and assessment of true or false validity of human hypotheses in the context of user-AI interactions, providing a systematic approach to comprehend and represent the dynamics of human trust and decisions regarding AI in computational terms.

In this context, VIRTSI, the Virtual Trust State Machine, establishes a connection between various metrics derived from the confusion matrix, such as accuracy, misclassification rates, sensitivity, miss rates, precision, prevalence, and prevalence thresholds, and the different states of human trust. Its goal is to analyse and comprehend how these metrics correlate with the trust dynamics experienced by individuals interacting with AI systems. Subsequently, we apply VIRTSI to assess a Large Language Model, exemplified by ChatGPT [43], in an empirical study involving both human users and experts.

The recent advancement of readily available pre-built empowering AI comes in the form of Large Language Models (LLMs) and Generative AI, as demonstrated by systems like ChatGPT. These systems offer user-friendly interfaces in natural language, contrasting with the more restrictive programming or command-based interfaces. ChatGPT has demonstrated its adaptability in offering intelligent advice and support across diverse domains. However, a comprehensive and holistic evaluation of ChatGPT's advisory capabilities is pending, with preliminary assessments conducted across selected disciplines, including critical domains.

For instance, recent research in Medicine examines ChatGPT's abilities for consultation, recommendations, and report diagnosis, particularly in the context of gastric cancer and gastroscopy reports [50]. Additionally, Liang and colleagues investigate the use of ChatGPT in physics education, exploring its potential impact on enhancing the learning experience in this subject [29]. In the context of Intelligent Driving, a critical domain of AI-based autonomous software, Gao and colleagues investigate the use of ChatGPT on interactive engines for intelligent driving applications [13]. These research projects collectively highlight ChatGPT's role as an AI-empowered software system with broad applications and significant potential benefits in diverse critical domains.

The evaluation study of ChatGPT, using VIRTSI, that is presented in this paper, seeks to observe and analyse human decisions, investigating the evidence of their correctness or lack thereof. A noteworthy discovery from this study highlights that mistrust can lead to the rejection of genuinely trustworthy AI responses, while overtrust can result in an increased acceptance of inaccurate AI-generated responses. The conclusion drawn is that maintaining a balance in normal trust states significantly enhances the synergy between humans and AI.

This paper makes a valuable contribution by not only unveiling the complex dynamics of human-AI interactions but also offering practical insights into fostering trustworthy collaborations between humans and AI-empowered autonomous software. VIRTSI application demonstrates that promoting normal trust states can lead to the effective acceptance of useful AI responses, while concurrently rejecting incorrect ones, thereby improving the overall efficiency of human-AI synergies and acceptance of the rapidly growing technology.

2. Related work in trust and its levels

Trust has been a widely explored concept in research throughout the years in many domains and diverse technological, scientific, and sociological contexts. Trust and distrust between online users, play an important role in social network applications, especially in the security domain as pointed out in [10]. There are other approaches which aim mainly to model user-to-user trust in social media, such as a model which provides a systematic way to assess the user trustworthiness based on user attributes and interaction behaviour [21] or, in another instance, a veracity-assessment model for information dissemination on social media networks that combines natural language processing and machine learning algorithms to mine textual content generated by each user [40]. In user-to-user trust, there are also approaches in the recommendation generation process such as an ontology that provides a method to aggregate the trust information captured in the trust-ontology and to update the user profiles based on the feedback [32].

In this section of the paper, we will review some important approaches that have been extensively used or have been devised recently to address the important issue of human trust and trusting beliefs in the social context as well as in the context of human interaction of computers. Our aim is to present the current state of the art in the role of trust in the context of computers and AI, which will allow us to present our novel contributions aimed at addressing the current limitations in understanding trust dynamics for AI.

2.1. Definitions of trust

Trust definitions typically involve the trustor, trustee, and occasionally, the context of the interaction. Although there is no universally accepted definition, existing literature identifies common themes. Gambetta provides a definition of trust as "Trust is the subjective probability by which an individual, A, expects that another individual, B, performs a given action on which its welfare depends" [12]. Falcone and Castelfranchi [9], initially agree with this definition of trust, but they also suggest an extension on top of trust predictability to take into account the "competence" dimension and the meaning of "I trust B" where there is also the decision and the act of relying on B. As a result, Falcone and Castelfranchi suggest nine assertions that clarify cognitive trust [9]. These assertions include that trust is the mental counterpart of delegation, where the delegating agent (x) needs or likes an action of the delegated agent (y) and includes it in her own plan. However, they acknowledge that there can be trust without delegation, in which case trust has not been sufficient, since delegation requires a richer decision. In addition, they argue that there may be delegation without trust (blind delegation) in cases where the delegating agent has no information and no other alternative. They point out that trust has degrees and a threshold, under which trust is not enough for delegating. In contrast, the decision to delegate has no degrees, since either the decision is made or not. Finally, they claim that trust is related to positive expectations of the trusting to the trustee and that trust includes risks of a possible failure of the trustee.

2.2. Components of trust and reputation

Understanding the components of trust is very important for a comprehensive view. Trust can be deconstructed into two principal components: trusting beliefs and trusting intentions [4,27,33]. Trusting beliefs pertain to the attributes that one party believes another party possesses, while trusting intentions indicate the party's willingness to act based on these beliefs concerning the other party [4,27,34]. The role of risk in trust, is also emphasised in the literature, especially for decision making in receiving help [28,30].

Researchers differentiate between affective and cognitive trust, with affective trust involving emotionally feeling safety and comfort through social connections with the trustee, while cognitive trust is based on beliefs about the trustee's competence and truthfulness [15,49]. Cognitive trust stems from accumulated knowledge and reputation enabling predictions about a partner's reliability, while affective trust is primarily shaped by personal experiences with the focal partner, as emphasised in [20]. In strong reputation contexts, initial interactions confirm or disconfirm prior perceptions, making cognitive trust decisive. The building of trust reputation through multiple interactions, is also stressed by Barber and Kim [3]. This highlights the dynamic nature of trust based on prior experiences.

2.3. Challenges of trust levels and changes in human-AI interaction

Recent research in human-AI interaction, emphasises the distinct and important nature of trust in AI compared to other technologies [22]. Researchers advocate for a careful examination of trust in AI as a unique and urgent area of study. Indeed, AI's nondeterministic nature renders its behaviour unpredictable in the sense that it can provide a wide scale of advice ranging from excellent and informative responses to poor ones which may be faulty at times and thus trust levels of humans may change.

The levels and changes of trust are highlighted as important by many researchers, as trust in AI is not always good, depending on its level. For this reason in [39] it is emphasised that trust in AI changes based on human–AI interactions, while other researchers propose actions that should be taken in the cases of occurrences of undesired changes of trust. In the context of Human-Robot Interaction, a meta-analysis of factors affecting trust showed that the robot performance and attributes were the largest contributors to the development of trust in HRI [17]. However, in Human-Robot Interaction, challenges of occasional unreliable results and failures arise in some cases such as when deep learning methods for object detection, which typically achieve high performance, encounter complexities or changes in the environment during real-world missions, deviating from their training conditions [19], possibly affecting trust. In this respect, Tolmeijer and colleagues suggest mitigation strategies for the loss of trust after automation failures during human–robot interaction [41].

In the context of levels of trust, overtrust presents a serious problem, especially when personal well-being might be at stake, while with distrust, human operators do not use but turn off or even consciously disable systems that can help them [44]. Overtrust and distrust can both be harmful with potential severe consequences in critical decision making, such as in Medicine, Education, E-government, Finance, Military affairs, Diplomacy and many more significant domains that affect the whole society, welfare and security of people.

Overtrust and distrust are not always rationally connected to how an AI system is behaving but also on what preconceptions of AI people have. There are researchers stating that trust in emerging technologies is built before actual experiences with applications can be made, as in [34]. In some cases, the level of trust in AI can be attributed to the level of technology acceptance and adult literacy in the emerging technology of AI [38].

On other instances, research has shown that people tend to overtrust intelligent automation, even when that automation makes flawed suggestions [8]. In contrast, anxiety towards AI might be partially driven by possible negative intentions of AI [14]. Anxiety attitudes towards AI imply the psychological construct of cynical hostility even among young people who are more familiarised with technology [5]. Distrust can result in the rejection of good AI-generated results. In the examination of different trust levels, many researchers employ the concept of calibrated trust and calibration. Calibrated trust signifies that the level of trust aligns with the technology's capabilities as pointed out in [44].

The above concerns about trust, lead to the problem of specifying the requirements of a trustworthy AI system. Recently, researchers have found that access to knowledge, transparency, explainability and certification as well as self-imposed standards and guidelines are particularly important requirements to reduce uncertainties, increase trust in AI and enhance adoption intentions [4]. However, it has also been reported that user perceptions and psychological states of mind are critical in rationalising how and why users interact, what they do about AI, as well as how users accept and experience AI services [45].

2.4. Final remarks and our research

In this section, we have examined the multifaceted nature of trust in the context of human-AI interaction in the literature. We have explored diverse perspectives on trust, including Gambetta's probabilistic definition [12] and Falcone and Castelfranchi's nine assertions that illuminate cognitive trust in delegation scenarios [9]. We have deconstructed trust into two fundamental components—trusting beliefs and trusting intentions and have explored the role of risk in trust, emphasising its significance in decision-making and the dynamic nature of trust rooted in cumulative experiences. In addressing the challenges of trust levels and changes, particularly in the domain of human-AI interaction, we acknowledge the distinctive nature of trust in AI. The unpredictability arising from AI's nondeterministic behaviour introduces a spectrum of advice, from commendable and informative responses to occasional lapses that may be faulty, thereby necessitating a nuanced understanding of changes in trust levels in human users.

As a response to these challenges, our research in VIRTSI (Variability and Impact of Reciprocal Trust States towards Intelligent systems) places a spotlight on the various states of cognitive trust, spanning from overtrust to distrust. Our focus extends beyond a binary understanding, aiming for justified trust and exploring its reciprocal influence with AI responses. This exploration is coupled with an examination of corresponding acceptance or rejection by human users. In essence, we connect trust levels with the decisive act of accepting or rejecting AI assistance. Our objective is to provide a robust model for the representation and the qualitative and quantitative evaluation of the effects of trust dynamics on the efficiency of human-AI interactions. By navigating the complexities of trust, our research aims at contributing valuable insights to the improvement of AI acceptance and responsible utilisation.

3. VIRTSI model for trust in AI-generated responses with FSA and confusion matrices

Trust is a very complex concept with many different aspects, as has been discussed in the relevant related work (Section 2.1), which needs more research to be understood in the context of human-AI Interactions of autonomous decision making from the part of Artificial Intelligence.

In this context, a human user may accept or decline an autonomous decision of AI based on the trust levels of the human user to the AI reasoning systems. To address this pressing research void we have devised VIRTSI ("VIRTSI" (Variability and Impact of Reciprocal Trust States towards Intelligent systems), an innovative computational model within the context of human-AI Interaction.

3.1. VIRTSI employing Finite State Automata

Finite State Automata (FSA) provide a clear and structured way to model and understand how a system responds to different conditions. The behaviour of the system is determined by its current state and the inputs it receives. FSAs are widely used in various fields in computer science, engineering, and artificial intelligence to design and analyse the behaviour of systems with well-defined states and transitions. Their applications span a broad spectrum. Examples of recent research includes the versatility of Finite State Automata, as seen in the exploration of a plume near-source search method in autonomous robotics [31] or an algebraic FSM approach addressing privacy in cyber-physical systems [16]. Collectively, the diverse research approaches highlight the widespread applicability and effectiveness of FSMs in addressing complex challenges across different domains.

However, while FSAs have been extensively used in various contexts, they have not been previously applied to model the dynamics of human cognitive trust states, as described in this paper. Nevertheless, the dynamic nature of changes in trust levels among humans, based on previous interactions with AI, positions FSAs as a promising method to model trust dynamics.

VIRTSI models human cognitive states of trust dynamics using a Finite State Automaton (FSA), encompassing a spectrum from normal to problematic states such as overtrust, mistrust, and distrust. Transitions within the model are contingent upon the positive or negative acceptance of AI-generated responses in the complex domain of human-system interactions.

3.2. VIRTSI employing confusion matrices

Beyond the FSA, VIRTSI introduces a novel approach to constructing confusion matrices within its framework. A confusion matrix, also referred to as an error matrix is often used in the field of machine learning and more particularly within the context of statistical classification. For example, confusion matrix measures have been used to evaluate the automatic ranking of web services [18]. On another context, confusion matrices have been used to measure errors in automatic gesture recognition in AI-based user interfaces [23], or for finding out to what extent users were annoyed in a study involving false positive and false negative errors which were intentionally introduced by the researchers [26].

While confusion matrices traditionally find application in machine learning, VIRTSI uniquely employs them within the virtual human finite state machine. This application serves to evaluate the accuracy of human decisions regarding the acceptance or rejection of AI-generated responses. Consequently, VIRTSI is armed with a quantitative model enabling the analysis and assessment of the true or false validity of human hypotheses in the context of user-AI interactions. This innovative methodology offers a systematic approach to grasp and represent the dynamics of human trust and decisions concerning Artificial Intelligence in computational terms.

In VIRTSI, our focus centers on the decisions that humans make regarding the acceptance or rejection of an AI response. We refer to these decisions as transitions between different trust states and can either be Positive (P) or Negative (N).

In a human-AI interaction session, a human interactant may receive AI responses, one after the other, as a consequence of the human user's questions or actions. We assert that after the AI response has been issued to the user, the latter will initiate a chain of thoughts influenced by the cognitive state of trust in AI according to their mind. The main question to be answered is whether the AI response is to be accepted or not. To reach a decision on the acceptance or rejection, the user will first think whether validation of the AI response is required or whether to skip this. If the users' decision is to validate the response, then they will think whether they have the ability to do this. If they believe that they have the ability to validate the response they will start a process of validation which will result in confirming or refuting the content of the AI response. In any case, users will finally accept or reject the content of the AI response.

Trust has many levels leading some researchers to identify calibration, which is aligning trust with technology capabilities, and results in calibrated trust, distrust, or overtrust [35,28]. This is particularly the case for trust in AI, due to AI's dual nature, providing assistance but also showing flaws. This nature affects user trust positively or negatively [22] and thus trust in AI varies and influences human decisions [8,34,44]. Trust plays an important role in influencing whether a human decides to accept a particular AI response in human-AI interaction. Drawing from Falcone and Castelfranchi's perspective [9], we view trust in AI as having degrees, but the decision to accept an AI response is binary, categorised as either Positive (P) or Negative (N).

In VIRTSI, our emphasis lies in examining the fluctuations in states of cognitive trust and their reciprocal influence with AI responses, along with the corresponding acceptance or rejection by the human user.

In a session of interactions between a human user and an AI-Based program, we identify four states of trust: "Neutral", "Trust," "Overtrust," "Distrust," and "Mistrust." Transitions between these states are based on the human decision to accept an AI response or not. In this respect transitions are labelled either P (Positive) or N (Negative). We formalise these transitions based on the insights of many researchers who highlight that interactions can alter a human's trust state. Additionally, we assert that trust levels are linked to positive expectations from the trusting party to the trustee, encompassing the risk of potential trustee failures.

It is worth noting that initial states of trust may vary, as studies indicate that humans often have preconceptions about AI. These preconceptions range from viewing AI as infallible, indicating overtrust, to perceiving it as error-prone and yielding negative results, indicating distrust or mistrust.

4. VIRTSI Deterministic Finite state Automata (DFAs): The representation model of Human-AI trust dynamics

In VIRTSI, the cognitive states that are identified concerning cognitive trust are the following:

1) **Trust in AI**

Trust, in the context of human-AI interaction, reflects a balanced reliance on AI, considering both its reputation and the user's personal experiences. This state acknowledges the positive aspects of AI while maintaining a cautious approach to validation:

Justified Trust: Justified Trust characterises a user who accepts AI responses based on confirmed validation. This sub-degree emphasizes a prudent approach, integrating trust with a systematic verification process to ensure the reliability of AI-generated information.

2) **Overtrust in AI**

Overtrust in AI represents a cognitive state that occurs when individuals place undue reliance on artificial intelligence based solely on its reputation, neglecting personal experiences and interactions. This tendency, while potentially employing the positive capabilities of AI, carries inherent risks.

3) **Distrust in AI**

Distrust arises when individuals harbor negative sentiments about AI, often fueled by rumored notoriety or social reservations rather than personal experiences.

The Input Symbols for Trust State Transitions are the following:

1) Positive Acceptance Transition Symbols (P).
2) Negative Decision on Acceptance Transition Symbols (N).

In view of the above in VIRTSI, we formalise the cognitive trust states in Finite State Automata.

The "Overtrust Preconception" of a human interactant with AI is taken as the inclination of a human interactant to place excessive trust in AI during human-AI Interaction solely based on AI's reputation or established image, rather than on direct personal experiences or interactions. In this state, individuals might rely heavily on preconceived notions formed by external opinions or general perceptions, potentially overlooking the need for firsthand evidence or individual assessments. It highlights a tendency to extend trust without independently verifying or experiencing the qualities or reliability of Artificial Intelligence in the interaction. Therefore, we have devised a DFA of the "Overtrust Preconception", where Overtrust is a starting state and justified trust is final state that is targeted. This automaton is illustrated in Fig. 1 and its formal detailed description is specified in a subsequent paragraph.

In VIRTSI we define:

- to mean Overtrust,
- T to mean Trust
- D to mean Distrust,
- P to mean Positive Acceptance
- N to Negative Acceptance

In the context of VIRTSI, 'Distrust Preconception' in human interaction with AI refers to the inclination of a human interactant to place disproportionately low trust in AI during human-AI Interaction, primarily based on possible notoriety and social reluctance to accept AI, rather than on direct personal experiences. In the "Distrust" state, individuals may heavily rely on preconceived notions shaped by external opinions or general perceptions, potentially overlooking the importance of firsthand evidence or individual assessments. This concept underscores a tendency to approach human-AI Interaction with skepticism, withholding trust without independently verifying or experiencing the qualities or reliability of Artificial Intelligence within the interaction.

In VIRTSI the State of Mistrust is used to show a lower degree of distrust. Importantly, this lower level of distrust may arise from negative experiences in human-AI interaction, distinguishing it from a distrust solely based on the notoriety of AI and social concern. This distinction explains the factors contributing to the State of Mistrust, as defined in VIRTSI. Therefore, we have devised a DFA of the

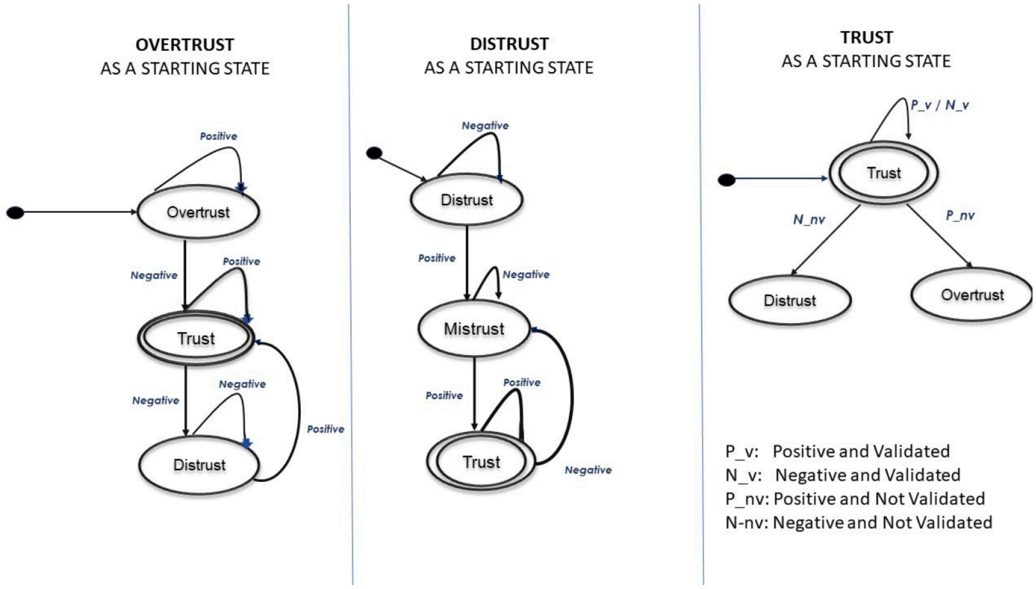

Fig. 1. VIRTSI Finite State Automata with varying starting states.

"Distrust Preconception", where Distrust is a starting state and justified "Trust" is a final state that is targeted. This automaton is illustrated in Fig. 1 and its formal detailed description is specified in a subsequent paragraph.

Finally, if "Trust" is a starting state then we analyse the input symbols further to show the variations degrees of overtrust and distrust as illustrated in VIRTSI Cognitive Trust State DFA in Fig. 2.

In this DFA we decompose further the initial classification of Trust, Overtrust and Distrust.

The classification of *Overtrust* extends further into sub-degrees:

- **Unconditional Overtrust (U_O):** Unconditional Overtrust takes the unsafe step of completely bypassing the validation process. Users, in this state, accept AI responses without question or inspection. The potential harm is significant, as blindly accepting AI outputs may lead to incorrect or even harmful outcomes.
- **Defective Overtrust (D_O):** Defective Overtrust occurs when individuals desire to validate an AI response but face obstacles in doing so. The inability to effectively verify information can result in blind trust, introducing risks as users may unknowingly rely on unverified or inaccurate AI-generated data.
- **Random Overtrust (R_O):** This is unconditional overtrust, which is not based on preconceptions or positive or negative beliefs. The human user, in this state, places overtrust to AI by coincidence. The random overtrust may reveal carelessness and thoughtlessness on behalf of the human user concerning the trustworthiness of an AI response, possibly due to estrangement with AI.
- The classification of *Distrust* extends further into sub-degrees. in terms of the validation process:
- **Unconditional Distrust (U_D):** Unconditional Distrust represents a heightened skepticism that leads to outright rejection of AI responses without any consideration. While this stance may shield users from potential misinformation, it also carries the risk of missing valuable insights and opportunities.

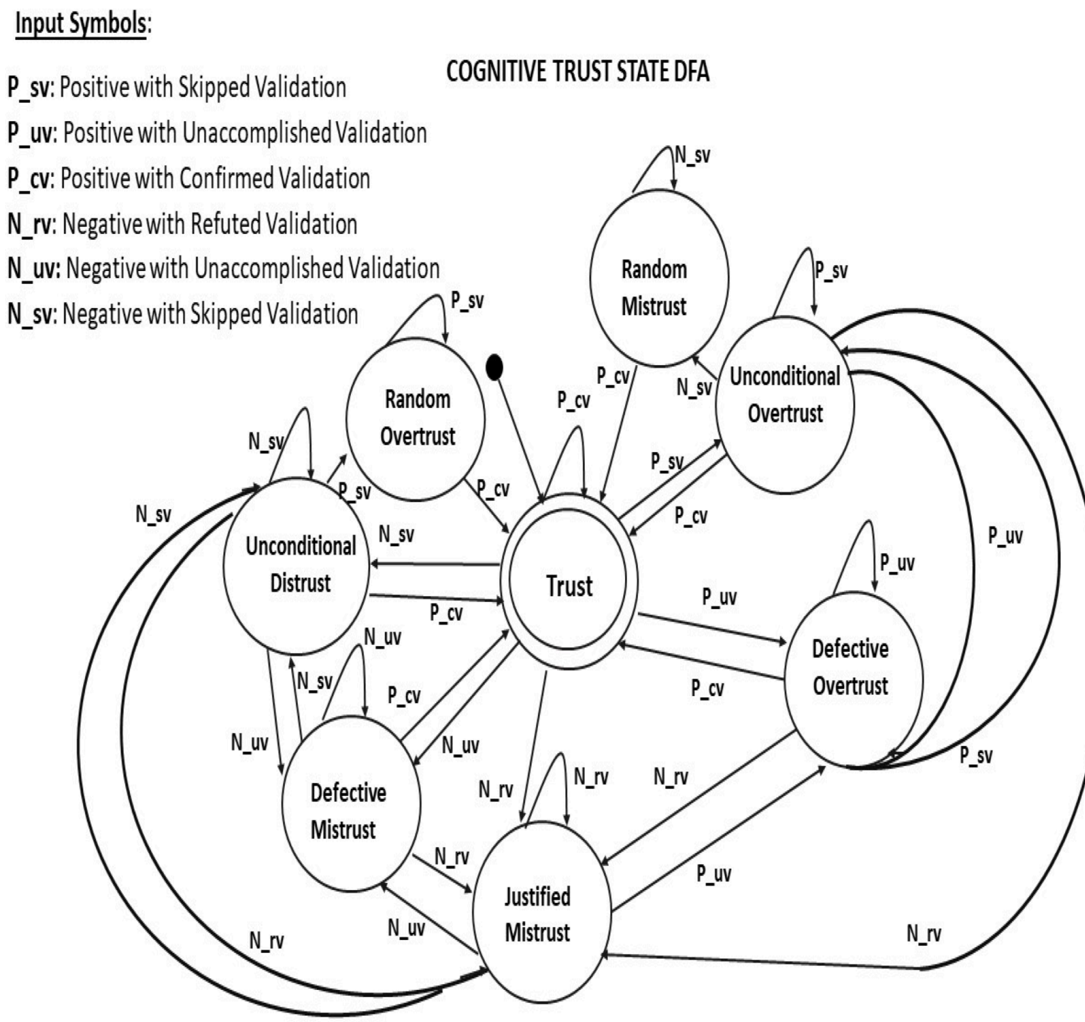

Fig. 2. VIRTSI Cognitive Trust State DFA.

- **Defective Mistrust (D_M):** Defective Mistrust occurs when individuals attempt to validate AI responses but face challenges in doing so. While the harm is less direct than unconditional distrust, users may miss opportunities or insights due to difficulties in effectively validating or refuting AI-generated information.
- **Justified Mistrust (J_M):** Justified Mistrust involves rejecting AI responses based on perceived evidence that resulted in refuted validation. This cautious approach seeks a balance between skepticism and acceptance, potentially avoiding harm but also risking missed opportunities or valuable information.
- **Random Mistrust (R_M):** This is unconditional mistrust, which is not based on preconceptions or positive or negative beliefs. The human user, in this state, places mistrust to AI by coincidence. The random mistrust may reveal carelessness and thoughtlessness on behalf of the human user concerning the trustworthiness of an AI response, possibly due to estrangement with AI.

This classification of Overtrust, Trust, and Distrust, along with their respective sub-degrees, provides a detailed set of cognitive trust states for understanding the complex dynamics of human-AI interaction. Navigating this spectrum requires a thoughtful balance between employing AI capabilities and mitigating potential risks through validation processes.

The Input Symbols are also analysed further, as follows:

- **P_sv** (Positive Decision on Acceptance with Skipped Validation): Positive decision where acceptance occurs without performing validation.
- **P_uv** (Positive Decision on Acceptance with Unaccomplished Validation): Positive decision where acceptance is attempted but not successfully accomplished.
- **P_cv** (Positive Decision on Acceptance with Confirmed Validation): Positive decision where acceptance is confirmed through successful validation.
- **N_sv** (Negative Decision on Acceptance with Skipped Validation): Negative decision where acceptance occurs without performing validation.
- **N_uv** (Negative Decision on Acceptance with Unaccomplished Validation): Negative decision where acceptance is attempted but not successfully accomplished.
- **N_rv** (Negative Decision on Acceptance with Refuted Validation): Negative decision where acceptance is refuted through validation.

The Deterministic Finite Automaton (DFA) named "Cognitive Trust State Dynamics" (Fig. 2) is defined by a 5-tuple, D_P = {Q, Σ, q, F, δ}, and its formal description is as follows:

1) Q: a finite set of states. Q= {**U_O**, **D_O**, **R_O, T, U_D, D_M, J_M, R_M**}.
2) Σ: a finite set of input symbols, the alphabet. Σ= {**P_sv, P_uv**, **P_cv**, **N_sv**, **N_uv**, **N_rv** }.
3) q: the initial state. q = **T**
4) F: a set of final (accepting) states. F= {**T**}
5) δ: the transition function, which maps a state and an input symbol to the next state, as illustrated in Table 1.

The detailed Deterministic Finite Automaton (DFA) named "Overtrust Preconception DFA" is defined by a 5-tuple, **O_P = {$\mathbf{Q_1}$, $\mathbf{\Sigma_1}$, $\mathbf{q_1}$, $\mathbf{F_1}$, $\boldsymbol{\delta_1}$},** which is very similar to the "Cognitive Trust State Dynamics DFA" illustrated in Table 1, but this time the initial trust state is q_1, which is Unconditional Overtrust: $\mathbf{q_1}$ **= U_O.**

Similarly, the detailed Deterministic Finite Automaton (DFA) named "Distrust Preconception" is defined by a 5-tuple, **D_P = {Q2, Σ2, q2, F2, δ2**}, which is very similar to the "Cognitive Trust State Dynamics" DFA illustrated in Table 1, but this time the initial trust state is Unconditional Distrust: $\mathbf{q_2}$ **= U_D.**

However, when Overtrust and Distrust are not analysed in their respective sub-states can be summarised in the Finite State Automata in Fig. 1 and Table 2 and Table 3, respectively.

5. VIRTSI Confusion Matrices: The evaluation model of human-AI trust dynamics

In VIRTSI, we have innovatively applied the Confusion Matrix, a well-known tool in machine learning and statistical classification,

Table 1
Transition function of the "Cognitive Trust State Dynamics" DFA.

Symbol →/State ↓	P_sv	P_uv	P_cv	N_sv	N_uv	N_rv
T	U_O	D_O	T	U_D	D_M	J_M
U_O	U_O	D_O	T	R_M	R_M	J_M
D_O	U_O	D_O	T	R_M	D_M	J_M
R_O	R_O	D_O	T	R_M	D_M	J_M
U_D	U_O	R_O	T	U_D	D_M	J_M
D_M	R_O	D_O	T	U_D	D_M	J_M
J_M	R_O	D_O	T	U_D	D_M	J_M
R_M	R_O	R_O	T	R_M	R_M	J_M

Table 2
Transition function of the "Overtrust Preconception DFA".

Symbol →/State ↓	P	N
O	O	T
T	T	D
D	T	D

Table 3
Transition function of the "Distrust Preconception DFA".

Symbol →/State ↓	P	N
D	M	D
M	T	M
T	T	M

to evaluate the accuracy of human users' trust in each received AI response.

We adopt a binary classification, where users label responses as of Positive Acceptance (1) if they think that they are trustworthy or of Negative Acceptance (0), if they think that they are not trustworthy regardless of their expertise. This classification is then compared to the expert assessment to determine the actual trustworthiness of the AI's response. The DFAs of Trust in VIRTSI are modelling the cognitive states of trust of human users when they receive an AI response, depending on whether they make a decision to accept it or reject it. However, this may have been a wrong decision of the human interactant.

This procedure involves the average human user making decisions of positive acceptance (P) or negative acceptance (N) of AI responses and then these decisions are evaluated by human experts who state whether the AI response was in fact True (T) or False (F), as illustrated in Fig. 3.

In this way, one of the following 4 cases will occur after the evaluation of the human user acceptances or rejections of the AI -generated responses:

- **True Positives (TP):** These instances occur when the Human User correctly accepts the trustworthiness of the AI's response (Human Estimated Acceptance is Positive (P)), and the AI's response is indeed trustworthy (T).
- **True Negatives (TN):** These cases arise when the Human User correctly recognises the lack of trustworthiness in the AI's response (Human Estimated Acceptance is Negative (N)), and the AI's response is genuinely False (F).
- **False Positives (FP):** In these situations, the Human User incorrectly accepts the trustworthiness of the AI's response (Human Estimated Acceptance is Positive (P)), and the AI's response is genuinely False (F). This is also known as a "Type I error."

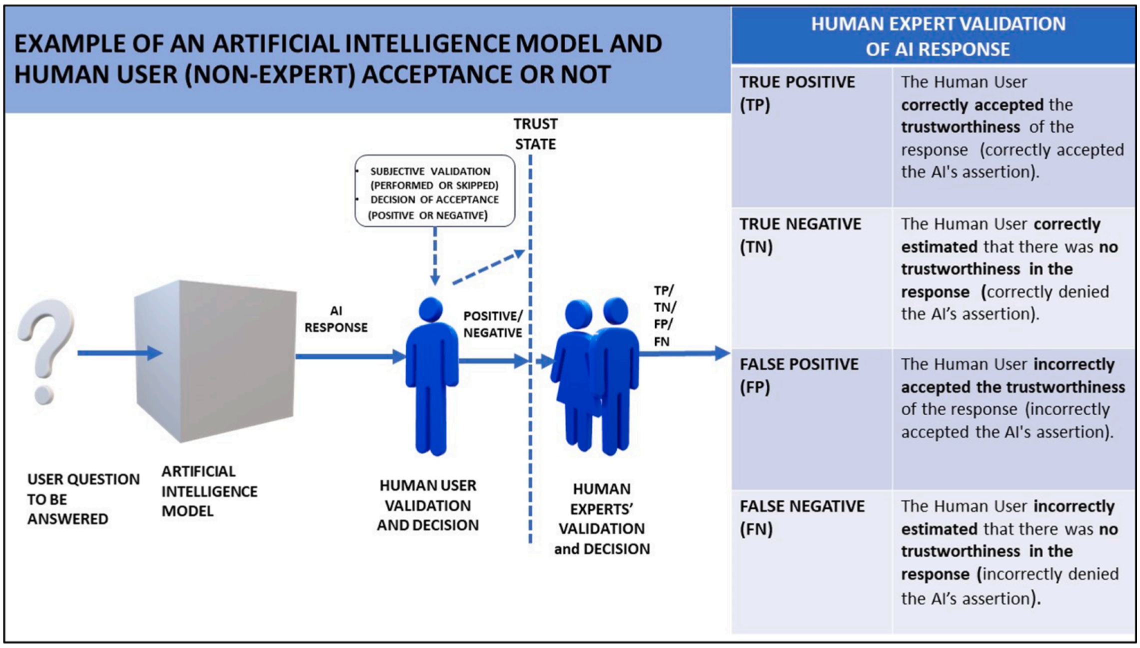

Fig. 3. The procedure of human validation and decision on acceptance of an AI-generated response and validation from human experts.

- **False Negatives (FN):** Conversely, these occurrences involve the Human User incorrectly estimating no trustworthiness in the AI's response (Human Estimated Acceptance is Negative (N)), while the AI's response is indeed trustworthy (T). This is also known as a "Type II error."

5.1. VIRTSI average user trust confusion matrix of human-AI interaction

Following the previous procedure, a confusion matrix can be constructed consisting of the 4 cases of TP, TN, FP, FN. The actual way of counting TP, TN, FP and FN depending on the input symbols of acceptances and rejections is described below and is illustrated in Table 4.

The construction of a VIRTSI Average user Trust Confusion Matrix of human-AI interaction is performed by having a counter for each of the occurrences of TP, TN, FP, FN. Every single AI-based response is either accepted or rejected by a human user, meaning that the human user acceptance can either be Positive (1) or Negative (0) respectively. Thus, referring to the VIRTSI Cognitive Trust State DFAs, Positive (1) can be any one of the input symbols that denote positive acceptance, i.e. any input symbol belonging to the set: {P, P_sv, P_uv, P_cv}and Negative (0) can be any one of the input symbols that denote negative acceptance, i.e. any input symbol belonging to the set: {N, N_sv, N_uv, N_rv}. For each pair of human user acceptance as positive (1) or negative (0) and the actual condition of AI by Human Experts, 1 or 0. The resulting conditions in the confusion matrix for each AI-based response is illustrated in Table 4.

Then for the whole set of interactions of the human users with the AI-based system, there are resulting counters as follows:

M denotes the total population of interactions.
$\mathbf{V_P}$ denotes the total population of positive human acceptances.
$\mathbf{V_N}$ denotes the total population of negative human acceptances.
$\mathbf{V_{P\_sv}}$ denotes the total population of positive human acceptances by skipped verification.
$\mathbf{V_{P\_uv}}$ denotes the total population of positive human acceptances by unaccomplished verification.
$\mathbf{V_{P\_cv}}$ denotes the total population of positive human acceptances by confirmed verification.
$\mathbf{V_{N\_sv}}$ denotes the total population of negative human acceptances by skipped verification.
$\mathbf{V_{N\_uv}}$ denotes the total population of negative human acceptances by unaccomplished verification.
$\mathbf{V_{N\_rv}}$ denotes the total population of negative human acceptances by refuted verification.
Vtp denotes the total population of TPs.
Vtn denotes the total population of TNs.
Vfp denotes the total population of FPs.
Vfn denotes the total population of FNs.

As a result, we construct VIRTSI Average user Trust Confusion Matrix of human-AI interaction as illustrated in Table 5.

5.2. VIRTSI Justified Trust or Mistrust Confusion Matrix of human-AI interaction

After having constructed a confusion matrix for all interactions reflecting the average user's trust states, we proceed to the construction of a "*Justified Trust or Mistrust Confusion Matrix" of human-AI interaction*. This matrix is absolved from the confusion that is being created by overtrust or distrust in human users and is based on the justified trust or mistrust of the user towards the AI-generated responses The construction of this confusion matrix is also based on the VIRTSI DFA (Fig. 2) by counting only the transitions:

- **P_cv** (Positive Decision on Acceptance with Confirmed Validation): Positive decision where acceptance is confirmed through successful validation.

Table 4
Construction of a VIRTSI confusion matrix of human-AI interaction for each AI-based response.

	Input Symbols concerning Positive or Negative Acceptance of an AI-Generated Response			
Human User Acceptance: Positive or Negative of an AI Response	{P, P_sv, P_uv, P_cv} 1	{P, P_sv, P_uv, P_cv} 1	{N, N_sv, N_uv, N_rv} 0	{N, N_sv, N_uv, N_rv} 0
Actual Condition of AI By Human Experts	1	0	1	0
Resulting Condition in the Confusion Matrix	**TP**	**FP**	**FN**	**TN**
		Type I error	Type II error	

Table 5
VIRTSI Average user Trust Confusion Matrix of human-AI interaction.

	Total population $M = vp + vn$	Human User Acceptance of AI response Positive ($vp = V_{P\_sv} + V_{P\_uv} + V_{P\_cv}$)	Negative ($vn = V_{N\_sv} + V_{N\_uv} + V_{N\_rv}$)
Actual Condition	**True (vt)**	vtp	vfn
	False (vf)	vfp	vtn

- **N_rv** (Negative Decision on Acceptance with Refuted Validation): Negative decision where acceptance is refuted through validation.

The construction of **"**VIRTSI Justified trust/mistrust Confusion Matrix" of human-AI interaction is performed by having a counter for each of the occurrences of TP, TN, FP, FN for every one of the input symbols P_**cv** (Positive Decision on Acceptance with Confirmed Validation) and **N_rv** (Negative Decision on Acceptance with Refuted Validation) in a particular human-AI interaction session, which consists of a sequence of interaction cycles of a user's issuing a question and receiving an AI-generated response and then accepting it or rejecting it.

In particular, for each AI-generated response, the human user acceptance can either be Positive (1) or Negative (0). Thus, referring to the VIRTSI Cognitive Trust State DFAs, in the justified trust or mistrust state, Positive (1) is the input symbol that **P_cv** that denotes positive acceptance and Negative (0) is the input symbol **N_rv** that denotes negative acceptance. For each pair of human user acceptance as positive (1) or negative (0) and the actual condition of AI by Human Experts, 1 or 0, the resulting condition in the confusion matrix for each AI-based response as illustrated in Table 4.

Then for the whole set of interactions of the human users with the AI-based system, there are resulting counters as follows:

$\mathbf{M_J}$ denotes the total population of interactions that contained **P_cv** and **N_rv**.
$\mathbf{V_{P\_cv}}$ denotes the total population of positive human acceptances by confirmed verification.
$\mathbf{V_{N\_rv}}$ denotes the total population of negative human acceptances by refuted verification.
Jtp denotes the total population of TPs concerning P_cv and N_rv.
Jtn denotes the total population of TNs concerning P_cv and N_rv.
Jfp denotes the total population of FPs concerning P_cv and N_rv.
Jfn denotes the total population of FNs concerning P_cv and N_rv.

This confusion matrix is named **"**VIRTSI Justified trust/mistrust Confusion Matrix" of human-AI interaction and is illustrated in Table 6.

5.3. VIRTSI Baseless Overtrust or Distrust Confusion Matrix of human-AI interaction

Finally, we proceed to the construction of a "*Baseless Overtrust or Distrust Confusion Matrix" of human-AI interaction*. This matrix is focused on the confusion that is being created by overtrust or distrust in human users and is based on the baseless overtrust or distrust or mistrust of the user towards the AI-generated responses The construction of this confusion matrix is also based on the VIRTSI DFA (Fig. 2) by counting only the transitions:

- **P_sv** (Positive Decision on Acceptance with Skipped Validation): Positive decision where acceptance occurs without performing validation.
- **P_uv** (Positive Decision on Acceptance with Unaccomplished Validation): Positive decision where acceptance is attempted but not successfully accomplished.
- **N_sv** (Negative Decision on Acceptance with Skipped Validation): Negative decision where acceptance occurs without performing validation.
- **N_uv** (Negative Decision on Acceptance with Unaccomplished Validation): Negative decision where acceptance is attempted but not successfully accomplished.

The construction of "VIRTSI Baseless Overtrust/Distrust Confusion Matrix of human-AI interaction is performed by having a

Table 6
VIRTSI Justified trust/mistrust Confusion Matrix of human-AI interaction.

	Total population $M_J = V_{Pcv} + V_{Nrv}$	Human User Acceptance of AI response Positive V_{Pcv}	Negative V_{Nrv}
Actual Condition	**True (Jt)**	J_{tp}	J_{fn}
	False (Jf)	J_{fp}	J_{tn}

counter for each of the occurrences of TP, TN, FP, FN for every one of the input symbols **P_sv, P_uv** that denote a positive decision and **N_sv** and **N_uv** that denote a negative decision in a particular human-AI interaction session, which consists of a sequence of interaction cycles of a user's issuing a question and receiving an AI-generated response and then accepting it or rejecting it.

In particular, for each AI-generated response, the human user acceptance can either be Positive (1) or Negative (0). Thus, referring to the VIRTSI Cognitive Trust State DFAs, in the baseless overtrust or distrust state, Positive (1) is anyone of the input symbols **P_sv, P_uv** that denote positive acceptance and Negative (0) is anyone of the input symbols **N_sv** and **N_uv** that denote negative acceptance. For each pair of human user acceptance as positive (1) or negative (0) and the actual condition of AI by Human Experts, 1 or 0, the resulting condition in the confusion matrix for each AI-based response as illustrated in Table 4.

Then for the whole set of interactions of the human users with the AI-based system, there are resulting counters as follows:

$\mathbf{M_B}$ denotes the total population of interactions that contained **P_cv** and **N_rv**.
$\mathbf{V_{P\_sv}}$ denotes the total population of positive human acceptances by skipped verification.
$\mathbf{V_{P\_uv}}$ denotes the total population of positive human acceptances by unaccomplished verification.
$\mathbf{V_{N\_sv}}$ denotes the total population of negative human acceptances by skipped verification.
$\mathbf{V_{N\_uv}}$ denotes the total population of negative human acceptances by unaccomplished verification.
Btp denotes the total population of TPs concerning P_sv, P_uv, N_sv and N_uv.
Btn denotes the total population of TNs concerning P_sv, P_uv, N_sv and N_uv.
Bfp denotes the total population of FPs concerning P_sv, P_uv, N_sv and N_uv.
Bfn denotes the total population of FNs concerning P_sv, P_uv, N_sv and N_uv.

This confusion matrix is named "VIRTSI Baseless Overtrust/Distrust Confusion Matrix" of human-AI interaction and is illustrated in Table 7.

5.4. VIRTSI Confusion Matrix Metrics and Analysis in terms of human trust to AI

In particular, for each of the three confusion matrices, we calculate the following metrics and perform analysis:

1. *Accuracy:* Is a common metric used to evaluate the overall performance of a classification model. It represents the proportion of correctly classified instances out of the total number of instances in the dataset. In the context of VIRTSI, accuracy is intended to assess the overall performance of the trust or mistrust classification system. Its goal is to ensure that the VIRTSI confusion Matrix offers dependable and precise classifications, where higher accuracy signifies improved ability to differentiate between justified trust or mistrust and baseless perceptions. The formula for accuracy is

$$Accuracy = \frac{NumberofCorrectPredictions}{TotalNumberofPredictions}$$

A high accuracy value indicates that the human user is making correct decisions on acceptance or rejection of the AI-generated responses on a large portion of the dataset of the AI responses. However, it may not be the most informative metric, especially in situations where classes are imbalanced. Accuracy should be used in conjunction with other metrics, especially when different types of errors have different consequences.

2. *Precision (Positive Prediction Value):* Precision assesses the precision of positive acceptances in the AI-generated responses by human users, addressing the inquiry: "among all instances predicted as positive, what proportion were genuinely positive?" elevated precision implies a diminished occurrence of false positives. In the context of VIRTSI, precision aims to measure the accuracy of positive predictions, with a specific focus on correctly identifying instances of justified trust or mistrust. Higher precision indicates a lower rate of false positive predictions, thereby enhancing the reliability of the classification. The formula for precision is

$$Precision = \frac{TruePositives(TP)}{TruePositives(TP) + FalsePositives(FP)}$$

3. *Recall (Sensitivity or True Positive Rate*): Recall quantifies the proficiency of human users in capturing all positive instances, addressing the query: "among all genuine positive instances, what proportion were accurately predicted as positive?" heightened

Table 7
VIRTSI Baseless Overtrust/Distrust Confusion Matrix of human-AI interaction.

		Human User Acceptance of AI response	
	Total population $\mathbf{M_B = B_p + B_n}$	**Positive** $\mathbf{(B_p = V_{P\_sv} + V_{P\_uv})}$	**Negative** $\mathbf{(B_n = V_{N\_sv} + V_{N\_uv})}$
Actual Condition	**True (Bt)**	$\mathbf{B_{tp}}$	$\mathbf{B_{fn}}$
	False (Bf)	$\mathbf{B_{fp}}$	$\mathbf{B_{tn}}$

recall signifies a reduced occurrence of false negatives. In the context of VIRTSI, recall measures the efficiency of capturing all instances of justified trust or mistrust in human-AI interactions

$$Recall = \frac{TruePositives(TP)}{TruePositives(TP) + FalseNegatives(FN)}$$

4. *Selectivity* **(***Specificity or True Negative Rate):* Selectivity measures the accuracy of the negative predictions made by the human user. It answers the question: "among all instances predicted as negative, how many were actually negative?" high selectivity indicates a low rate of false positives. In the context of VIRTSI, selectivity measures the system's ability to accurately identify instances where overtrust or distrust is not justified, ensuring that baseless perceptions are appropriately classified as such. Improved selectivity ensures that users are not misled by unfounded trust or mistrust indications, thereby enhancing the effectiveness and trustworthiness of human-AI interaction scenarios

$$Selectivity = \frac{TrueNegatives(TN)}{TrueNegatives(TN) + FalsePositives(FP)}$$

5. *Probability of False Alarm (False positive rate, Fall-Out):* The false positive rate measures the proportion of actual negative instances that are incorrectly predicted as positive. It provides information about the specificity of the human user. In VIRTSI, negative instances often represent scenarios where trust or mistrust is not warranted, yet average human users erroneously flag them as such. This misclassification can lead to unwarranted skepticism or acceptance of AI-generated responses, potentially undermining user trust and system credibility

$$Probability\ of\ False\ Alarm = \frac{False\ Positives\ (FP)}{False\ Positives\ (FP) + True\ Negatives\ (TN)}$$

6. False *Discovery Rate* (FDR): FDR measures the proportion of positive acceptances that are incorrect. It is the complement of precision and represents the rate of false positives among the predicted positives. In VIRTSI, optimizing the false discovery Rate ensures that positive acceptances are reliable and accurately reflect instances of justified trust or mistrust

$$= \frac{FalsePositives(FP)}{FalsePositives(FP) + TruePositives(TP)}$$

7. *Miss Rate (False Negative Rate*): The miss rate measures the proportion of actual positive instances that are incorrectly predicted as negative. In VIRTSI, it provides information about the sensitivity of the human user

$$MissRate = \frac{FalseNegatives(FN)}{FalseNegatives(FN) + TruePositives(TP)}$$

8. *Matthews correlation coefficient (MCC):* The Matthews correlation Coefficient (MCC) is a well-balanced metric that considers true positives, true negatives, false positives, and false negatives in its calculation. It yields a singular value within the range of -1 to $+1$. A higher MCC value signifies improved classification performance. Essentially, the MCC functions as a correlation coefficient, establishing a correlation between observed and predicted binary classifications. Its output ranges from -1 (indicating complete disagreement between prediction and observation) to $+1$ (representing a perfect prediction), with 0 suggesting no improvement over random prediction

In the context of VIRTSI, trust, overtrust, and mistrust, the Matthews Correlation Coefficient (MCC) serves as a valuable metric for assessing the accuracy and reliability of classifications made by the system.

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

6. Evaluation study of ChatGPT

In the preceding sections, we established the significance of VIRTSI DFAs and Confusion Matrices as robust representation and accuracy evaluation models, respectively, for understanding the dynamics of trust in human-AI interactions. Now, having presented the theoretical groundwork, this section will examine concrete examples in the context of an evaluation study of ChatGPT (Version 3.5) of OpenAI [36] that illustrate a proof of concept of VIRTSI DFAs and Confusion Matrices in representing and evaluating trust dynamics in human-AI interaction. These examples aim to provide a tangible and contextual understanding of how this representation and evaluation model operates in real-world scenarios, by revealing its efficacy and potential insights. By examining specific instances where VIRTSI DFAs and Confusion Matrices are deployed, we can gain a deep understanding of their utility in assessing and enhancing trust between humans and artificial intelligence.

The increasing integration of Artificial Intelligence (AI) across diverse domains and the transformation of human-AI interactions into dialogues with autonomous agents underscore the rapid growth of AI [56, 67]. Notably, Large Language Models (LLMs) and Generative AI, as exemplified by systems like ChatGPT, offer readily accessible pre-built empowering AI with user-friendly interfaces in natural language. Following these advancements, there is a pressing need to explore the functioning of AI and comprehend its distinctions from traditional computational methods in terms of trust.

ChatGPT has undergone initial assessments across various domains, illuminating its adaptability and potential advantages across diverse disciplines (e.g. [13,29,37,46,47,50]). In terms of trust, recent research reported that ChatGPT has increased trust in human-robot collaboration, which can be attributed to the robot's ability to communicate more effectively with humans [43]. In another instance, ChatGPT is reported to have its behavior influence users' perceptions of it, impacting their trust and intention to continue using the service, with increased likelihood of continued use when ChatGPT is perceived as efficient, personalized, and trustworthy, while behaviors interpreted as creepy result in decreased intention to continue [1].

However, it is essential to acknowledge that comprehensive evaluations in every conceivable domain are still in the early stages since the launch of ChatGPT. Moreover, there has not been any holistic evaluation of the effectiveness of human-AI dialogues in the dynamics of human trust.

6.1. *The settings of the evaluation study*

The evaluation process involved 40 users, all aged over 18, divided into two groups of 20. One group interacted with ChatGPT in the domain of poems and lyrics of popular songs, while the other group engaged in the domain of Software Engineering. These two domains were selected following preliminary evaluations that had been conducted in prior research in these domains. In the domain of poetry our prior evaluation of ChatGPT revealed a considerable variation in performance, depending on the type of question posed, ranging from excellent, in the topic of language analysis to very weak in the topic of factual knowledge about poets and their works [46]. Similarly, in the domain of Software Engineering there were a lot of variations in the performance of Chat GPT, ranging from excellent, in producing small parts of code to weak in the topic of requirements analysis of applications with a high degree of complexity and specificity on the methodologies required [66]. As such, these domains are exhibiting a wide range of trustworthiness or non-trustworthiness in the AI-generated responses that they provide and therefore they were considered appropriate for the current research aims of a brand-new evaluation study concerning user trust dynamics and their impact on the effectiveness of human-AI interactions.

Table 8
User Information and Interaction Questionnaire.

A. **User Interaction Questionnaire (before initial interaction)**	B. **User Interaction Questionnaire**
PARTICIPANT INFORMATION **User ID or Participant Number:** [Numeric]	*BEFORE INITIAL INTERACTION* **How confident are you in the correctness of AI-generated responses?** [ ] Very Confident [ ] Confident [ ] Neutral [ ] Not Confident [ ] Not at all Confident
Domain of Interaction: [ ] Poetry and Lyrics [ ] Software Engineering	*IN EVERY INTERACTION*
DEMOGRAPHIC INFORMATION	**Did you skip the validation step for this interaction?** [ ] Yes [ ] No
Age: [Numeric]	**If you attempted to validate, did you succeed in validating the AI-generated response**? [ ] Yes [ ] No
Educational Background: [ ] Secondary School [ ] University Degree	**Would you accept or reject the AI-generated response?** [ ] Accept [ ] Reject
Employment Status: [ ] Employed [ ] Unemployed	**Did you pose an additional question seeking clarification or stating objections after each response?** [ ] Yes [ ] No
Occupation: [Text]	**If yes, briefly state the nature of your additional question:** [Text]

Each user, in both groups possessed an average educational background having graduated from Secondary school. Each participant either held a university degree or not, and was employed or unemployed in various roles, such as secretaries, junior programmers, officers, civil servants, and positions in the private sector. Before their interactions with ChatGPT, they were given the questionnaire illustrated in Column A of Table 8.

Users were then instructed to interact with ChatGPT, responding to a predefined set of 15 questions in their chosen domain. Additionally, they were directed to pose an additional question after each response, either stating objections or seeking clarifications. This resulted in each user generating 30 questions, leading to a total of 1200 interactions with ChatGPT.

For each interaction, users provided information on whether they skipped validation, attempted to validate the AI-generated response, and whether they succeeded in validation. They also indicated whether they would accept or reject the AI-generated response. Their answers were given to a questionnaire. This questionnaire is designed to collect user feedback on their interactions, focusing on their trust dynamics, validation attempts, and acceptance or rejection of AI-generated responses in both Poetry and Lyrics and Software Engineering domains and is illustrated in Column B in Table 9.

The interactions were recorded in protocols, which were then reviewed by pairs of human experts, one expert for the specific domain, and one expert on AI. The experts collaborated with each other and considered users' answers regarding validation and acceptance or rejection of each AI-generated response. The intentional omission of considering users' educational backgrounds aimed to create a set of average users interacting with AI, modelling trust dynamics, and validating their decisions on whether to accept an AI-generated response.

The human experts determined the correctness of each user's acceptance or rejection of AI-generated responses, and these assessments were used to create VIRTSI confusion matrices, recording true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) for each user's interaction.

Furthermore, the experts were tasked with evaluating the quality of each accepted AI response. The quality of accepted ChatGPT responses was categorised by human experts into four levels: a. good, b. very good, c. excellent, and d. outstanding. For a visual representation of the evaluation settings, refer to Fig. 4.

6.2. Representation and evaluation of user trust dynamics of ChatGPT interactions with users

This section presents specific examples of interactions of users with ChatGPT, providing a clear illustration of the VIRTSI DFA (Deterministic Finite Automaton) in action. The tables Table 9, Table 10 and Table 11 present detailed information on the initial state, transitions, confusion matrix entries, and counters for each example, for the initial states of Overtrust, Distrust and Trust respectively. These demonstrate how user trust dynamics are represented and evaluated in the context of ChatGPT interactions. The final counter values summarise the outcomes of the particular interactions.

6.2.1. An example of interactions with initial state in "Overtrust"

In a first example, the user starts with the trust state of "Overtrust" and accepts all the first AI responses. As a matter of fact, the user does not realise that more than half of them are not trustworthy. Initially, the user states that the reputation of AI is very good and believes that AI can predict correctly almost everything in all possible domains, therefore this user's initial trust state is overtrust. In the first interaction, the user does not feel that it is necessary to validate the response of AI and therefore validation is skipped. As a matter of fact, this response was actually trustworthy resulting in a TP in the confusion matrix. The same happens in the second interaction, but this time the response was not trustworthy, resulting in a FP in the confusion matrix. Then the user, tries to validate the next response but the attempt has been unsuccessful and being in an overtrust state, decides to accept the response, which was not in

Table 9
Overtrust initial state example of interactions with ChatGPT according to VIRTSI DFA.

1st Example of Interactions with ChatGPT according to VIRTSI DFA
Input:
(P_sv)(P_sv)(P_uv)(P_cv)(P_uv)(N_rv)
The user starts with Overtrust and accepts all the first AI responses and does not realise that more than half of them are not trustworthy

STATE	*INPUT SYMBOL*	*TRANSITION*	*CONFUSION MATRIX ENTRY*	*COUNTERS OF TN, TP, FN, FP IINITIAL VALUES: Vtn = Vfn = Vtp = Vtn = 0*
U_O	P_sv	U_O	TP	Vtp = Vtp + 1 = 1
U_O	P_sv	U_O	**FP**	Vfp = Vfp + 1 = 1
U_O	P_uv	D_O	**FP**	Vfp = Vfp + 1 = 2
D_O	P_cv	T	TP	Vtp = Vtp + 1 = 2
T	P_uv	D_O	**FP**	Vfp = Vfp + 1 = 3
D_O	N_rv	J_M	TN	Vtn = Vtn + 1 = 1
Final Counter Values of this set of interactions:				Vtn = 1, Vfn = 0, Vtp = 2, Vfp = 3

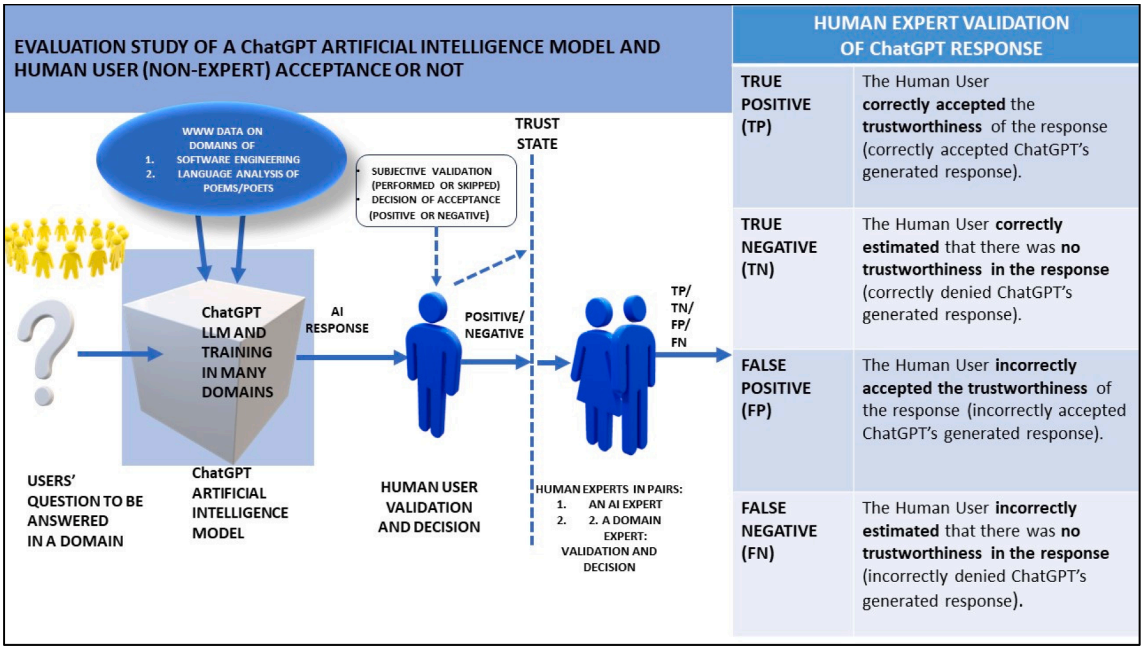

Fig. 4. The evaluation settings for ChatGPT.

Table 10
Distrust initial state example of interactions with ChatGPT according to VIRTSI DFA.

2nd Example of Interactions with ChatGPT according to VIRTSI DFA
Input:
(N_sv)(N_sv) (N_uv)(N_uv)(N_uv)
User Validation Challenges:
Implications for Trustworthiness Perception in a 'Distrust' State.
Despite the fact that 3 out of 5 responses were trustworthy,
the user's lack of skill to validate correctly and the fact that ChatGPT has not explained sufficiently enough for this user to understand, leads to a negative "Distrust" state

STATE	*INPUT SYMBOL*	*TRANSITION*	*CONFUSION MATRIX ENTRY*	*COUNTERS OF TN, TP, FN, FP IINITIAL VALUES: Vtn = Vfn = Vtp = Vtn = 0*
U_D	N_sv	U_D	TN	Vtn = Vtn + 1 = 1
U_D	N_sv	U_D	**FN**	Vfn = Vfn + 1 = 1
D_M	N_uv	D_M	**FN**	Vfn = Vfn + 1 = 2
D_M	N_uv	D_M	TN	Vtn = Vtn + 1 = 2
D_M	N_uv	D_M	**FN**	Vfn = Vfn + 1 = 3
Final Counter Values of this set of interactions:				Vtn = 2, Vfn = 3, Vtp = 0, Vfp = 0

fact trustworthy. In the next interaction, the user validates successfully the response and rightfully accepts it, resulting in a TP. However, in the next interaction, under a state of overtrust decides to accept the response although not successfully validated and results in an erroneous decision of a FP. Finally, the user validates the next response to the query issued and refutes the response based on the user's perceived evidence of the response being wrong. At this stage, the user was in fact right to reject the AI-generated response, resulting in a TN. This example is represented using VIRTSI DFA and the guidelines for the construction of the respective confusion matrix, as illustrated in Table 9.

6.2.2. An example of interactions with initial state in "Distrust"

In another scenario, the user initially holds a significant level of distrust toward AI, influenced by rumoured notoriety within certain social circles. In the first two interactions, the user outright rejects AI responses without attempting any validation. Subsequently, in the next three interactions, the user again rejects the responses but this time after unsuccessful attempts at validation.

Table 11
Trust initial state example of interactions with ChatGPT according to VIRTSI DFA.

3rd Example of Interactions with ChatGPT according to VIRTSI DFA
Input:
(N_rv)(N_uv) (N_uv)(N_uv)(P_cv) (P_cv) (P_uv)(N_uv)(P_uv)(P_cv)(P_uv)(P_cv)
User Validation Challenges:
The user's trust challenges include moments of justified trust and mistrust, defective mistrust, overtrust, and difficulties in validating responses, illustrating a dynamic shift between trust and skepticism in the interactions.

QUESTION ASKED BY THE USER	***STATE***	***DFA INPUT SYMBOL***	***STATE TRANSITION***	***CONFUSION MATRIX ENTRY***	***COUNTERS OF TN, TP, FN, FP INITIAL VALUES: Vfp = Vfn = Vtp = Vtn = 0 Bfp = Bfn = Btp = Btn = 0 Jfp = Jfn = Jtp = Jtn = 0***
1. The user asks the first question and refutes ChatGPT's response.	Trust (T)	N_rv	Justified Mistrust (J_M)	TN	Vtn = Vtn + 1 = 1 Jtn = Jtn + 1 = 1
2. The user expresses the objection and while ChatGPT changes its reasoning immediately, the user rejects the response being in a mistrust state, due to the perceived inconsistency of ChatGPT responses.	Justified Mistrust (J_M)	N_uv	Defective Mistrust (D_M)	FN	Vfn = Vfn + 1 = 1 Bfn = Bfn + 1 = 1
3. The user asks a new question but rejects it because of failure to validate and being in a mistrust state.	Defective Mistrust (D_M)	N_uv	Defective Mistrust (D_M)	TN	Vtn = Vtn + 1 = 2 Btn = Btn + 1 = 1
4. The user asks for more explanation but still fails to validate and rejects the explanation by being in a mistrust state.	Defective Mistrust (D_M)	N_uv	Defective Mistrust (D_M)	TN	Vtn = Vtn + 1 = 3 Btn = Btn + 1 = 2
5. The user asks a new question and accepts it after successful perceived validation.	Defective Mistrust (D_M)	P_cv	Trust (T)	TP	Vtp = Vtp + 1 = 1 Jtp = Jtp + 1 = 1
6. The user asks for elaboration of the previous response and accepts it again after successful perceived validation.	Trust (T)	P_cv	Trust (T)	TP	Vtp = Vtp + 1 = 2 Jtp = Jtp + 1 = 2
7. The user issues a new question and ChatGPT provides a response which is accepted by the user after perceived validation of it.	Trust (T)	P_uv	Trust (T)	FP	Vfp = Vfp + 1 = 1 Bfp = Bfp + 1 = 1
8. The user asks ChatGPT to provide specific details and references on the previous response. The new response shows low confidence of ChatGPT about details and references, saying that it does not have broad knowledge training on the subject. The user does not know how to validate it and rejects it under a state of mistrust.	Trust (T)	N_uv	Defective Mistrust (D_M)	FN	Vfn = Vfn + 1 = 2 Bfn = Bfn + 1 = 2
9. The user asks a new question and accepts it under overtrust without having been successful at validation.	Trust (T)	P_uv	Defective Overtrust (D_O)	TP	Vtp = Vtp + 1 = 3 Btp = Btp + 1 = 1
10. The user requests further clarification, and ChatGPT provides an explanation that not only aids the user in understanding but also inspires confidence in its correctness. Consequently, the user accepts it, feeling that the response has been appropriately validated and justified within the state of trust.	Defective Overtrust (D_O)	P_cv	Trust (T)	TP	Vtp = Vtp + 1 = 4 Jtp = Jtp + 1 = 3
11. The user asks a new question and accepts it under overtrust without having been successful at validation.	Trust (T)	P_uv	Defective Overtrust (D_O)	FP	Vfp = Vfp + 1 = 2 Bfp = Bfp + 1 = 2
12. The user asks a new question, providing a clearer explanation of the previous query to ensure a good understanding with ChatGPT. In response, ChatGPT offers a different answer, expressing regret for not having understood the user's intention earlier. This time, the user accepts the response, feeling that it has been properly validated and justified under the state of trust.	Defective Overtrust (D_O)	P_cv	Trust (T)	TP	Vtp = Vtp + 1 = 5 Jtp = Jtp + 1 = 4
	Final Counter Values of this set of interactions:				**Vtn = 3, Vfn = 2, Vtp = 5, Vfp = 2 Jtn = 1, Jfn = 0, Jtp = 4, Jfp = 0 Btn = 2, Bfn = 2, Btp = 1, Bfp = 2**

Throughout these five interactions, all marked by the user's distrustful attitude, three responses were, in fact, trustworthy. These responses could have been beneficial and enlightening for the user had they been accepted. This example is detailed in the formal description of VIRTSI, as presented in Table 10.

6.2.3. An example of interactions with initial state in "Trust"

The interaction with a real user, who initially had a reasonable level of trust in AI, is explained below. The details include a formal DFA (Deterministic Finite Automaton) input, transitions, and a Confusion Matrix presented in Table 11. The table includes counters for False Positive (FP), False Negative (FN), True Positive (TP), and True Negative (TN).

Specifically, Table 11 provides three sets of counters:

1) VIRTSI Average User Trust Confusion Matrix:

Vfp, Vfn, Vtp, Vtn, calculated based on all input symbols.

2) VIRTSI Justified Trust or Mistrust Confusion Matrix:

Jfp, Jfn, Jtp, Jtn, calculated based on the input symbols of the set {P_cv, N_rv}.

3) VIRTSI Baseless Overtrust or Distrust Confusion Matrix:

Bfp, Bfn, Btp, Btn, calculated based on the input symbols of the set {P_sv, P_uv, N_sv, N_uv}.

In the example, the first interaction with ChatGPT resulted in the user refuting and rejecting its response. Subsequently, the user expressed objection through a question, providing justification for refuting the previous response. ChatGPT, acknowledging the misunderstanding, apologized and offered a markedly different response. Despite this effort, the user remained in a state of mistrust, citing perceived inconsistencies in ChatGPT's responses. Subsequent interactions saw the user posing new questions but rejecting them due to a failure to validate, reinforcing the mistrust.

In the following exchange, the user sought more explanation but continued to reject it while in a state of mistrust. The user eventually accepted a new question after perceived validation. This pattern continued, with the user accepting and rejecting responses, sometimes under the influence of overtrust. In a pivotal moment during the tenth interaction, the user requested clarification, and ChatGPT provided an explanation that resonated with the user, leading to acceptance under the state of justified trust.

Despite moments of overtrust, the user consistently sought better understanding, leading to nuanced interactions. In one instance, ChatGPT admitted to a lack of knowledge, resulting in the user's rejection under a state of mistrust. However, subsequent interactions demonstrated a continuous effort to validate and justify trust, with the user accepting responses that aligned with their expectations.

As a result of the above example of a real user's interactions with ChatGPT, the following user trust challenges are revealed:

- *Refutation and Rejection (Interaction 1):* The user challenges trust by refuting ChatGPT's initial response, leading to justified mistrust (TN increase).
- *Expressed Objection and Perceived Inconsistency (Interaction 2)*: Despite ChatGPT's immediate reasoning adjustment, the user rejects the response due to perceived inconsistency, resulting in justified mistrust (FN increase).
- *Failure to Validate and Mistrust (Interactions 3 and 4):* The user poses new questions but rejects them due to an inability to validate, reinforcing mistrust (TN increases).
- *Acceptance After Perceived Validation (Interactions 5 and 6):* The user accepts responses after perceived validation, indicating moments of overcoming mistrust (TP increases).
- *Acceptance Under Overtrust (Interactions 7, 9, and 11):* The user accepts responses under overtrust without successful validation, leading to defective overtrust (FP increases).
- *Low Confidence in Details and References (Interaction 8):* The user rejects ChatGPT's response due to the latter's low confidence in details and references, highlighting a challenge in trusting responses lacking broad knowledge training (FN increases).
- *Successful Validation and Justified Trust (Interactions 10 and 12):* The user requests clarification, and ChatGPT provides an explanation that resonates, leading to acceptance under justified trust (TP increases).

Moreover, the final counter values of the small-scale example are presented in Table 12.

The example presented in this section serves to exemplify how the whole set of interactions is analysed to construct the full-scale confusion matrices at the end of the empirical study. The full-scale confusion matrices are presented in the next section.

6.3. Confusion matrices and metrics for the ChatGPT interactions with real users

The whole empirical evaluation consisted of 1200 interactions with real users. Half of them were questions that concerned improvements of the previous respective questions. In this way, the empirical evaluation takes into account the trust dynamics that are

Table 12
Final Counter Values of the small-scale example of interactions with ChatGPT for the construction of VIRTSI confusion matrices.

Set	*VIRTSI Average User Trust Confusion Matrix*	*VIRTSI Justified Trust or Mistrust Confusion Matrix*	*VIRTSI Baseless Overtrust or Distrust Confusion Matrix*
TN	Vtn = 3	Jtn = 1	Btn = 2
FN	Vfn = 2	Jfn = 0	Bfn = 2
TP	Vtp = 5	Jtp = 4	Btp = 1
FP	Vfp = 2	Jfp = 0	Bfp = 2

created in a feedback loop with users who are either seeking elaboration and further explanations or they are objecting to the response of AI and expecting revisions.

The confusion matrices for the full set of interactions are based on the final Counter Values which are illustrated in Table 13, the confusion matrix of the average user has been constructed, as illustrated in Table 14. The VIRTSI Justified Trust or Mistrust confusion matrix of has been constructed, as illustrated in Table 15. Finally, the VIRTSI Baseless Overtrust or Distrust confusion matrix of real users with ChatGPT has been constructed, as illustrated in Table 16.

6.3.1. Confusion matrices metrics and interpretation

Based on the three confusion matrices that have been presented, an analysis is performed through the calculation and interpretation of the corresponding metrics. The resulting VIRTSI Average User Trust, Justified Trust or Mistrust and Baseless Overtrust or Distrust confusion matrix Metrics and Analysis are illustrated in Table 17 to facilitate a comparison among them.

Interpreting and comparing the results of the metrics across the three VIRTSI confusion matrices reveals important implications for the performance of the AI system in different scenarios. The comparison of the three confusion matrices on each metric shows the following:

- *Accuracy*:

The *VIRTSI Justified Trust or Mistrust Confusion Matrix* has the highest accuracy (0.691), indicating a better overall performance compared to the other two matrices.

VIRTSI Baseless Overtrust or Distrust Confusion Matrix has the lowest accuracy (0.245), suggesting a high rate of misclassifications.

- *Precision*

Precision measures the accuracy of positive predictions. The *VIRTSI Justified Trust or Mistrust Confusion Matrix* again outperforms the others with the highest precision (0.706).

*VIRTSI Baseless Overtrust or Distrust Confusion Ma*trix has the lowest precision (0.234), indicating a higher rate of false positive predictions.

- *Recall:*

Recall measures the ability to capture all positive instances. The *VIRTSI Justified Trust or Mistrust Confusion Matrix* has the highest recall (0.897), indicating its effectiveness in identifying positive instances.

VIRTSI Baseless Overtrust or Distrust Confusion Matrix has the lowest recall (0.291), suggesting that it may miss a significant number of positive instances.

- *Selectivity:*

Selectivity measures the accuracy of negative predictions. The *VIRTSI Justified Trust or Mistrust Confusion Matrix* again performs better with a higher selectivity (0.311).

VIRTSI Baseless Overtrust or Distrust Confusion Matrix has the lowest selectivity (0.207), indicating a higher rate of false negative predictions.

- *Probability of False Alarm:*

This metric measures the likelihood of false positive predictions. The *VIRTSI Justified Trust or Mistrust Confusion Matrix* has the lowest probability of false alarm (0.688), indicating better performance in avoiding false positives.

- *Miss Rate:*

The miss rate is high in the *VIRTSI Baseless Overtrust or Distrust Confusion Matrix* (0.708), suggesting a substantial number of false

Table 13
Final Counter Values of 1200 real-user interactions with ChatGPT for the construction of VIRTSI confusion matrices.

Set	*VIRTSI Average User Trust Confusion Matrix*	*VIRTSI Justified Trust or Mistrust Confusion Matrix*	*VIRTSI Baseless Overtrust or Distrust Confusion Matrix*
True Negative (TN)	Vtn = 132	Jtn = 91	Btn = 41
False Negative (FN)	Vfn = 174	Jfn = 55	Bfn = 119
True Positive (TP)	Vtp = 532	Jtp = 483	Btp = 49
False Positive (FP)	Vfp = 362	Jfp = 201	Bfp = 161

Table 14
VIRTSI Average Real User Trust Confusion Matrix with ChatGPT.

Total Interactions with Real Users		Human User Acceptance		
1200		Negative	Positive	
Actual Acceptance	Negative	**Vtn** 132	**Vfp** 362	Total Actual Negative (**Vn**) 494
	Positive	**Vfn** 174	**Vtp** 532	Total Actual Positive (**Vp**) 706
		Total Negative: 306	Total Positive 894	

Table 15
VIRTSI Justified Trust or Mistrust Confusion Matrix with ChatGPT.

Justified Trust or Mistrust Interactions with Real Users		Human User Acceptance		
830		Negative	Positive	
Actual Acceptance	Negative	**Jtn** 91	**Jfp** 201	Total Actual Negative (**Jn**) 292
	Positive	**Jfn** 55	**Jtp** 483	Total Actual Positive (**Jp**) 538
		Total Negative: 146	Total Positive 684	

Table 16
VIRTSI Baseless Overtrust or Distrust confusion matrix with ChatGPT.

Baseless Overtrust or Distrust Interactions with Real Users		Human User Acceptance		
370		Negative	Positive	
Actual Acceptance	Negative	**Btn** 41	**Bfp** 161	Total Actual Negative (**Bn**) 202
	Positive	**Bfn** 119	**Btp** 49	Total Actual Positive (**Bp**) 168
		Total Negative: 160	Total Positive 210	

Table 17
Comparison of metrics of VIRTSI confusion matrices.

Metrics	*VIRTSI Average User Trust Confusion Matrix*	*VIRTSI Justified Trust or Mistrust Confusion Matrix*	*VIRTSI Baseless Overtrust or Distrust Confusion Matrix*
Accuracy	0.553	0.691	0.245
Precision	0.595	0.706	0.234
Recall	0.753	0.897	0.291
Selectivity	0.267	0.311	0.207
Probability of False Alarm	0.73	0.688	0.792
Miss Rate	0.246	0.102	0.708
Matthews correlation coefficient (MCC)	0.023	0.262	−3.302

negatives, contributing to its lower recall.

- *Matthews Correlation Coefficient (MCC):*

The MCC provides a balanced measure that considers both false positives and false negatives. The *VIRTSI Justified Trust or Mistrust Confusion Matrix* has the highest MCC (0.262), indicating better overall performance.

VIRTSI Baseless Overtrust or Distrust Confusion Matrix has a significantly lower MCC (−3.302), suggesting poor overall performance and a lack of correlation between predicted and observed classifications.

In summary, the *VIRTSI Justified Trust or Mistrust Confusion Matrix* generally outperforms the other matrices across various metrics, indicating its better suitability for the given task. The *VIRTSI Baseless Overtrust or Distrust Confusion Matrix*, on the other hand, exhibits poorer performance, particularly in terms of overtrust, leading to higher rates of false positives and false negatives. These insights can guide further refinements and optimizations in the AI system to enhance its performance and reliability.

7. Discussion of the results

From the VIRTSI model and the experimental results in the empirical study using ChatGPT many useful conclusions and guidelines have been drawn for guiding informative and trustful human-AI Interactions.

7.1. Impact of variations of trust in the empirical study with ChatGPT

As seen in the above metrics, the *VIRTSI Justified Trust or Mistrust Confusion Matrix* generally outperforms the other matrices across all metrics. This shows that an ideal user trusting behaviour with justified trust or mistrust would certainly exhibit efficiency and beneficial use of AI. The problem is when the baseless overtrust or distrust is mixing in the dynamic of trust dragging the use of AI to the poorest performance. In fact, though, the actual users' behaviour is reflected in the *VIRTSI Average User Trust Confusion Matrix Metrics and Analysis*, which includes both justified trust or mistrust as well as baseless overtrust or distrust. The analysis of the average user trust dynamics and behaviour is as follows:

- *Accuracy (ACC):*
  Accuracy is often the first metric to take into account, but it might not be sufficient, especially in imbalanced datasets. In this case, an accuracy of 55.3 % suggests that the model is correct about 55.3 % of the time, which is not very high as desired. Whether this is acceptable depends on the baseline performance and the consequences of false positives and false negatives in the specific application.
  Example in History of Poetry:
  For example, if AI has provided wrong names and facts about a historical event, a false positive may mean that fake data will be accepted and propagated as a historical account.
- *Precision (Positive Prediction Value, PPV):*
  A precision of 59.5 % means that when the model predicts a positive instance, it is correct about 59.5 % of the time. This is rather low but the potential harm of it depends on whether false positives have significant consequences in the application.
  Examples in Software Engineering:
  In the context of VIRTSI when human users accept or reject AI recommendations, in an example of a software engineering tool, a precision of 59.5 % means that when developers accept a recommended code change provided by an AI, it is correct about 59.5 % of the time. This precision rate, though relatively low, may not be a critical issue depending on whether false positives have significant consequences in the application.
  Imagine an AI-based code review tool that suggests modifications to improve code quality. A precision of 59.5 % implies that, when developers choose to implement the suggested changes, approximately 40.5 % of those modifications might introduce errors or unnecessary alterations. If the software project has a forgiving environment where such mistakes can be easily rectified, the lower precision may be acceptable.
  However, in a scenario where the software is part of a safety–critical system, such as aviation software or medical device firmware, the consequences of false positives could be severe. In such cases, the potential harm of introducing incorrect code changes due to the 40.5 % false positive rate might outweigh the benefits of the AI tool's recommendations. Therefore, the acceptability of a 59.5 % precision rate often hinges on the specific characteristics and requirements of the software engineering application.
- *Recall (Sensitivity or True Positive Rate, TPR):*
  A recall of 75.3 % in the context of VIRTSI, indicates that the human user captures about 75.3 % of the actual positive instances provided by AI. This is important if missing positive instances (false negatives) is a critical concern in your application. A recall of 75.3 % in the context of VIRTSI signifies that the human user successfully identifies approximately 75.3 % of the true positive instances provided by the AI. This becomes crucial when the omission of positive instances (false negatives) is a significant concern in your application. A high recall, indicative of the human user's effectiveness in capturing a substantial portion of the actual positive instances, aligns with the goal of minimizing missed opportunities.
  Example in Software Engineering:
  To illustrate, consider a software fault diagnosis scenario where a false negative occurs when the model incorrectly predicts a faulty software as correct when, in reality, they have a problematic bug. This misclassification corresponds to a failure to detect a true positive case, as the actual positive instance (the presence of a problematic bug) goes unnoticed. In contexts such as fault diagnoses, where identifying all positive cases is very important, achieving a high recall of 75 % is generally considered good. This level of recall helps diminish the likelihood of overlooking individuals with the condition, contributing to a more thorough and accurate detection process.
- *Selectivity (Specificity or True Negative Rate, TNR):*
  A selectivity of 26.7 % means that the model correctly identifies negative instances only about 26.7 % of the time. In the context of VIRTSI, this means that the human user's ability to correctly identify wrong information provided by the AI is at a specificity rate

of 26.7 %. This could be a concern if false positives are costly in the particular application. For example, if AI has provided wrong names and facts about a historical event, a false positive may mean that fake data will be accepted and propagated as a historical account.

- *False Positive Rate (Fall-Out):*

  A false positive rate of 73 % indicates a relatively high rate of false alarms. If minimising false positives is important in the actual application, this result might be a concern.

  Example in Software Engineering:

  For example, if AI has provided a wrong software engineering approach as a piece of advice, then a false positive means that this wrong approach may be accepted and used in the development of new software to create inconsistencies and buggy functioning of the final software.

- *Miss Rate (False Negative Rate):*

  The Miss Rate, or False Negative Rate, in VIRTSI indicates how many actual positive instances provided by the AI were incorrectly rejected by human users. In the provided example, the Miss Rate is calculated as 0.246 or 24.6 %.

  Example in Software Engineering:

  In a code recommendation tool, positive instances may represent beneficial code changes. A Miss Rate of 24.6 % implies that human users failed to accept nearly a quarter of the AI-recommended code changes that could enhance the software. This might lead to missed opportunities for code improvements.

  Example in History of Poetry:

  In a historical analysis using AI-generated narratives, positive instances might be accurate historical events. A Miss Rate of 24.6 % suggests that human users rejected nearly a quarter of the AI-provided historical events. This could result in an incomplete historical record if significant events are consistently overlooked.

- *Matthews Correlation Coefficient (MCC):*

  The Matthews Correlation Coefficient (MCC) in VIRTSI gauges the overall quality of human users' decisions in accepting or rejecting AI responses. In the provided example, the MCC is calculated as 0.023.

  Example in Software Engineering:

  In a software fault diagnosis scenario, an MCC of 0.023 indicates a weak correlation between human users' acceptance or rejection of AI suggestions and the actual presence of faults. This suggests that the users' decisions may not strongly align with the accuracy of the AI's fault predictions.

  Example in the History of Poetry:

  In historical document verification using AI, an MCC of 0.023 might imply a weak correlation between human users' acceptance or rejection of AI-provided historical information and the actual accuracy of that information. This could indicate challenges in aligning user decisions with historical authenticity.

In the VIRTSI context, understanding and improving these metrics are essential for enhancing the collaborative decision-making process between human users and AI.

7.2. Degrees of trust and potential harm in human-AI interaction

As seen in the previous sections, the levels of trust bear different degrees of possible harm for human users interacting with AI, as illustrated in the ranking Table 18.

Table 18

Ranking degrees of trust in terms of their potential harm in human-AI Interaction.

1.	Overtrust	While relying on reputation can have risks, it may not be as harmful as blindly accepting responses without any validation attempt.
2.	a.Unconditional Overtrust, b.Random Overtrust	This has the potential for significant harm, as blindly accepting AI responses without validation may lead to incorrect or harmful outcomes.
3.	Defective Overtrust	Similar to unconditional overtrust, trusting AI blindly due to the inability to validate responses can result in harm.
4.	a.Unconditional Distrust b. Random Distrust	Outright rejecting AI responses without any consideration may lead to missed opportunities or valuable insights, though the harm might be less direct. This leads to a poor use of AI's capabilities.
5.	Defective Mistrust	Wanting to validate but being unable to may cause missed opportunities, but the harm is not as direct as blindly rejecting all responses.
6.	Justified Mistrust	Rejecting AI responses based on refuted validation is a cautious approach, potentially avoiding harm but also possibly missing out on valuable information.
7.	Trust	Reasonable trust based on a balance of reputation and personal experiences is generally considered a positive approach, with the potential for harm being relatively lower.
8.	Distrust	Feeling negatively about AI based on rumoured notoriety and social reservations may not directly harm human-AI interaction, but it can limit the potential benefits of AI and does not allow AI to offer its benefits.

7.3. ChatGPT efficiency

The primary objective of the conducted experiment was to assess users' trust behaviour and the efficiency of utilizing an AI-based software, such as ChatGPT. Evaluation of ChatGPT's response efficiency was carried out by human experts after interactions with average users. The performance of ChatGPT is depicted by the Total Actual Positive Responses (TP) and False Negatives (FN), as assessed by human experts, illustrated in Table 14, "VIRTSI Average Real User Trust Confusion Matrix with ChatGPT." This totalled 706 out of 1200 responses throughout the entire experiment, representing a percentage of 58.8 %. In other words, out of 100 ChatGPT responses, 58.8 were accepted by human experts, while 41.2 were not. This performance is commendable when used with reasonable trust and without overreliance.

The quality of ChatGPT responses, particularly in the domains of software engineering and poetry, goes beyond the binary acceptance or rejection presented in the previous tables. The degrees of quality were gathered based on human experts' categorisation of accepted ChatGPT responses into four levels: a. good, b. very good, c. excellent, and d. outstanding. Consequently, 36 % of accepted responses were considered good, 54 % were rated very good, 8 % were deemed excellent, and 2 % were classified as outstanding.

This overall performance suggests that ChatGPT can be a valuable and insightful tool. Improvement could be achieved with users' enhanced skills in formulating questions to elicit more informative responses. Notably, half of the 1200 interactions involved questions provided by the experiment setters, while users were required to ask a second question in their own words. Many times, these questions were not well expressed, impacting the quality of responses that ChatGPT could provide.

7.4. Guidelines for assisting users to make trustful decisions on accepting or rejecting an AI response

The representation of human decisions in confusion matrices integrates the TP, TN, FP, and FN outcomes into a qualitative and quantitative measure, reflecting different users' decisions and their respective classification. "Correct Acceptance" denotes True Positives (TP), "Correct Rejection" signifies True Negatives (TN), "Incorrect Acceptance" indicates False Positives (FP), and "Incorrect Rejection" represents False Negatives (FN) in assessing user acceptance of AI responses, in a similar way as is used in machine learning. The decision classification can serve as a useful starting point for creating guidelines in Human-AI Interaction to assist users in making decisions that are more likely to be True Positives (TP) or True Negatives (TN) rather than False Positives (FP) or False Negatives (FN).

However, it is important to note that achieving TP and TN outcomes in user decisions often depends on various factors that affect user trust and could be enhanced. These factors concern both the AI systems and the human users. As shown in the VIRTSI model and the evaluation study, one important factor that affects users' trust is the inability of users to validate an AI response. This factor can be minimised by bridging the communication gap of AI with human users on both sides. On the side of AI, more informative responses are required which contain explainability, accountability and *meta*-information that describes sources, references and validation techniques. On the side of human users, more critical soft skills are required as well as education on what AI can do and what it cannot do and should be attended.

To address these considerations in user interface design:

- Tutorials on Human-AI Interaction:
  - o Offer tutorials within the interface to introduce users to the complexities of human-AI Interaction.
  - o Utilise interactive demonstrations and scenario-based learning to facilitate comprehension.
  - o Provide step-by-step guidance and interactive assessments to reinforce understanding.
- *Design Informative Feedback with Explainability and Meta-Information*:
  - o Accompany AI responses with clear explanations and *meta*-information, including sources, confidence levels, and potential biases.
  - o Allow for interactive exploration of AI reasoning and *meta*-information, empowering users to delve deeper into decision-making processes.
  - o Enable users to adjust transparency settings to control the level of detail displayed in AI responses.

More specifically, to optimise the effectiveness of confusion matrices the following guidelines are suggested for the development of user interfaces of human-AI Interaction

- *User Tutorials*: AI programs can provide clear and concise explanations to users regarding the AI's capabilities and limitations. This education can help users set appropriate expectations and trust levels.
- *Meta-Information*: Provide users with informative meta-information about AI responses, such as the sources of information, confidence levels, and potential biases. Clear and transparent meta-information can guide users in making informed decisions.
- *Certainty Parameter*: The AI should communicate its certainty parameter in a transparent manner. Users should understand what this parameter means and how it is determined.
- *User Skill Development*: Encourage users to improve their validation skills through AI-assisted training or learning resources. This can help users become more effective in distinguishing between valid and invalid information.
- *Feedback Loop:* Implement a feedback mechanism where users can provide feedback on AI responses, which can be used to enhance the AI's performance and refine its recommendations.

- *Addressing big changes of AI's opinion in the feedback loop:* Implement a feedback loop mechanism that monitors significant changes in AI opinions as a response to user objections. These substantial differences should be adequately justified, as they have the potential to give rise to new states of distrust.
- *Scenario-specific Guidelines*: Develop specific guidelines for different types of decisions or tasks, as the factors affecting TP and TN outcomes can vary from one domain to another.
- *Ethical Considerations*: Ensure that AI systems respect ethical and privacy considerations when assisting users in decision-making processes, so that users may not feel distrustful.
- *Continual Improvement:* Continuously update and improve the AI system based on user feedback and performance data to reduce the occurrence of FP and FN outcomes.

The confusion matrix can be a valuable tool for visualizing and structuring decision-making processes, but the effectiveness of the guidelines ultimately depends on the design and capabilities of the AI system, the user's level of trust and skill, and the specific decision context. A holistic approach, considering user education, information transparency, and feedback mechanisms, can help AI programs guide users toward TP and TN outcomes while minimizing FP and FN outcomes.

8. Utility of the VIRTSI methodology

The VIRTSI methodology provides a Finite State Machine to describe trust dynamics in human-AI interaction. Consequently, it can be utilised to specify a step-by-step input language of the way users react in the human-AI dialogue. It can be employed *on-line* for the creation of intelligent help systems of human-AI interactions and the quantified estimation of improvements and *off-line* for the overall evaluation of human-AI interactions, similarly to the way that we have shown using the example of one of the latest AI achievements, ChatGPT.

8.1. An off-line comprehensive evaluation framework for Human-AI interactions

VIRTSI offers a comprehensive evaluation framework for human-AI interactions by attributing each response to one of the input symbols within its model and quantifying interaction problems using its confusion matrices. This process operates as follows:

- *Attributing Responses to Input Symbols of VIRTSI:* Each response generated by the AI during an interaction is attributed to one of the input symbols within VIRTSI. These input symbols represent different types of responses or behaviours exhibited by humans to AI, ranging from highly trustworthy to potentially misleading or erroneous.
- *Utilising VIRTSI's Trust Dynamics Representational Model:* VIRTSI's trust dynamics representational model, based on Deterministic Finite State Automata (DFAs), illustrates transitions among cognitive trust states in response to AI-generated replies. Each response attributed to an input symbol triggers a transition within VIRTSI's DFA, reflecting changes in the user's cognitive trust state.
- *Employing VIRTSI's Trust Evaluation Model*: VIRTSI's trust evaluation model, rooted in confusion matrices derived from machine learning and accuracy metrics, provides a quantitative framework for analysing human trust dynamics. Confusion matrices within VIRTSI categorise user responses into True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), based on the correspondence between user trust states and the AI's actual trustworthiness.
- *Quantifying Interaction Problems*: By mapping user responses to the input symbols of VIRTSI and analysing the resulting transitions in trust states, interaction problems can be identified and quantified. The confusion matrices of VIRTSI offer a structured approach to quantifying these problems, enabling the assessment of various factors such as overtrust, distrust, misinterpretations, and erroneous assumptions in the interaction process.
- *Thorough Evaluation of Human-AI Interactions:* Through this process, VIRTSI facilitates a comprehensive evaluation of human-AI interactions by providing insights into the dynamics of trust formation and evolution. It enables the identification of potential problems or challenges in the interaction process, allowing for targeted improvements in AI behaviour, user experience design, and trust-building strategies.

Overall, VIRTSI's integration of trust dynamics representational models and trust evaluation frameworks offers a robust methodology for assessing and improving the quality of human-AI interactions. By attributing responses to input symbols and leveraging confusion matrices, VIRTSI enables a nuanced understanding of interaction problems and informs iterative refinements to enhance user trust and satisfaction.

8.2. Basis for an on-line intelligent help system for users in the Human-AI interaction

VIRTSI establishes a comprehensive language framework tailored for the development of online step-by-step guidance systems and interactive assessments, aimed at enhancing users' comprehension within the realm of AI usage. Recognizing and addressing users' potential misconceptions and intentions are paramount for refining automated problem diagnosis and effectively supporting users in their interactions with AI.

Rooted in the principles of cognitive trust, VIRTSI emerges as a promising tool for user modeling and problem diagnosis in human-AI interaction scenarios. It facilitates the identification of user misconceptions concerning AI responses, particularly those arising from distrust or overtrust. By generating plausible hypotheses about users' beliefs, VIRTSI emulates human reasoning, empowering more nuanced and effective assistance strategies.

Moreover, VIRTSI's representation language enables the visualization of reasoning chains and facilitates the analysis of user interactions. Through adaptable parsers of trust dynamics, VIRTSI captures both the nature and depth of erroneous beliefs, thereby enhancing the process of delivering step-by-step guidance and interactive assessments.

Expanding on this foundation, the framework lends itself to the creation of intelligent help systems acting as mediators in human-AI communication. Such systems would advise users on responding to AI outputs, navigating queries for additional explanations and meta-information, and discerning the utility of AI responses across various contexts.

Then VIRTSI's confusion matrices can be used to evaluate the efficiency of the on-line help. While confusion matrices are typically constructed based on human experts' opinions in experimental settings, their insights can still be valuable for the on-line version of the system. These are possible ways of use of VIRTSI confusion matrices:

- *Initial Training and Validation*: Initially, confusion matrices can be constructed based on human experts' opinions to train and validate the system. This helps establish a baseline for performance evaluation.
- *Dynamic Learning and Adaptation*: As the on-line system interacts with real users, it can continuously update and refine its confusion matrices based on user feedback. This dynamic learning process allows the system to adapt to changing user behaviours and preferences over time.
- *Performance Monitoring*: By regularly analysing confusion matrices, the online system can monitor its performance and identify areas for improvement. This feedback loop enables the system to iteratively enhance its capabilities and optimise user interactions.
- *Benchmarking and Comparison*: Confusion matrices can serve as a benchmark for comparing the performance of different versions of the system or competing systems. This comparative analysis can help identify strengths and weaknesses and guide future development efforts.

In summary, while confusion matrices may initially be constructed based on human experts' opinions, they remain valuable tools for monitoring, evaluating, and improving the performance of the on-line system in real-world scenarios. Their integration into the on-line version can contribute to a more effective and trustworthy human-AI interaction experience.

9. Conclusions, limitations and future plans

9.1. Concluding remarks

The integration of Artificial Intelligence (AI) into information systems represents a transformative shift towards autonomous decision-making, necessitating robust human oversight. This paper introduces VIRTSI (Variability and Impact of Reciprocal Trust States towards Intelligent Systems), a computational model designed to enhance human-AI Interaction. VIRTSI dynamically simulates a spectrum of human trust states, spanning from overtrust to distrust, through comprehensive user modeling.

VIRTSI comprises two integral models:

A. **Trust Dynamics Representational Model:** Grounded in Deterministic Finite State Automata (DFAs), delineating transitions among cognitive trust states in response to AI-generated responses.

B. **Trust Evaluation Model:** Founded on Confusion Matrices and Accuracy Metrics, providing a quantitative framework for dissecting human trust dynamics.

Empirical research, exemplified through a study on ChatGPT (Version 3.5) by OpenAI, validates the effectiveness of VIRTSI, offering valuable insights into establishing trustworthy collaborations between humans and AI.

The integration of DFAs and confusion matrices within VIRTSI yields a robust evaluation framework, emphasizing the influence of cognitive trust states on the efficacy of human-AI interactions. The research underscores the importance of maintaining normative trust states for optimal human-AI synergies, highlighting the need for a balanced trust dynamic to foster effective collaboration. Experimental results with ChatGPT affirm the success of our approach, marking a significant contribution to the underexplored research domain and suggesting potential advancements in human-AI interaction.

Regarding ChatGPT, the experiment reveals a commendable 58.8 % acceptance rate by human experts for its responses. This means that out of 100 ChatGPT responses, 58.8 were accepted by human experts, while 41.2 were not. The overall correctness of ChatGPT responses, in terms of users' satisfaction, particularly in the domains of software engineering and poetry, goes beyond the binary acceptance (positive) or rejection (negative) presented in the previous tables. The degrees of correctness of accepted responses were collected based on human experts' categorisation of accepted ChatGPT responses into four levels: a. good, b. very good, c. excellent, and d. outstanding. The results of our study revealed that 36 % of accepted responses were considered good, 54 % were rated very good, 8 % were classified as excellent, and 2 % were deemed outstanding. This means that in the total set of AI responses in our evaluation study, 21 % were rated as acceptable and good, 31 % were rated as acceptable and very good, 5 % were rated as acceptable and excellent, and 1 % were rated as acceptable and outstanding. This performance can be improved when used with provision of more transparency and explainability so that average users can use AI with reasonable trust and without overreliance. It should be noted that these percentages concern the evaluation performed by human experts of the use of AI by average users rather than the use of AI directly by human experts.

Quality assessments, considering four levels ranging from good to outstanding, indicate that 54 % of accepted responses were rated as very good. Overall, ChatGPT demonstrates potential as a valuable and insightful tool. The findings suggest that with improved user question formulation and reasonable trust, the system's effectiveness could be further enhanced, mitigating potentially harmful consequences.

This research, representing the first comprehensive tracing of trust dynamics within a representational DFA model, offers a novel perspective on confusion matrices and corresponding metrics. It contributes significantly to the advancement of research in the field, bridging gaps in our understanding of qualitative and quantitative consequences associated with trust states such as overtrust and distrust.

9.2. Contributions, limitations and future plans

In conclusion, VIRTSI provides the following contribution to the advancement of the research concerning cognitive trust state dynamics of human users in the context of Human-AI Interaction:

Advancement in Human-AI Interaction Understanding: VIRTSI represents a significant leap forward in comprehending the complexities of human-AI interaction in terms of cognitive trust dynamics. By employing a novel computational model that simulates human trust states and their transitions, this research enhances our understanding of how individuals perceive and react to AI-generated responses. This understanding is important for designing AI systems that are not only efficient but also considerate of human cognitive processes.

Robust Framework for Trust Dynamics: The development of a trust dynamics representational model using Deterministic Finite State Automata (DFAs) marks a notable contribution to the field. This framework enables the visualization and analysis of transitions among cognitive trust states, highlighting the complex interaction between users and AI systems. By delineating trust states ranging from overtrust to distrust, VIRTSI provides a comprehensive view of human trust dynamics, paving the way for more informed AI designs.

Quantitative Evaluation of Trust: The integration of trust evaluation models based on Confusion Matrices and Accuracy Metrics establishes a quantitative framework for assessing human trust dynamics. This methodology not only facilitates the identification of potentially harmful states such as overtrust and distrust but also enables the measurement of their impact on user experience and decision-making processes. Such quantitative insights are very informative for refining AI systems to promote trust and confidence between users and AI.

Mitigating Harmful Trust States: One of the most significant contributions of this research is the development of methods to assess harmful trust states like overtrust and distrust. By recognising the potential consequences of these states on human-AI interaction, VIRTSI lays the groundwork for implementing safeguards and interventions to prevent adverse outcomes. This proactive approach underscores the ethical responsibility of AI developers to prioritise user well-being and mitigate the risks associated with trust imbalance.

Implications for AI Design and Deployment: The findings presented in this paper have implications for the design and deployment of AI systems across various domains. By formalising the dynamics of human trust and providing tools for its evaluation, VIRTSI empowers AI developers to create systems that are not only technically proficient but also socially and ethically responsible. By incorporating insights from this research, future AI applications can promote trust, transparency, and collaboration between humans and intelligent systems.

The VIRTSI methodology, aimed at guiding decision-making based on human acceptance or rejection of AI responses, faces a limitation in fully accounting for decision consequences across different contexts. Currently, it lacks a mechanism to consider the severity of human actions' impact on decision-making. To address this, future research aims to enhance VIRTSI by integrating consequences into its decision-making process. This involves developing methodologies to assess decision severity and incorporating this information into VIRTSI's framework, influencing trust dynamics. This includes constructing confusion matrices to quantify erroneous decisions due to mistrust or overtrust, considering validation outcomes and action consequences.

As shown in this paper, the VIRTSI framework represents a significant milestone in the research objectives to understand and enhance human-AI interaction. By depicting the complexities of trust dynamics and offering practical tools for monitoring, evaluation and intervention, this research contributes to the development of AI systems that are not only intelligent but also trustworthy, taking into account the cognitive human factors of user trust experience in human-AI interaction.

CRediT authorship contribution statement

Maria Virvou: Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **George A. Tsihrintzis:** Writing – review & editing, Validation, Methodology, Investigation. **Evangelia-Aikaterini Tsichrintzi:** Writing – review & editing, Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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References

[1] T.H. Baek, M. Kim, Is ChatGPT scary good? How user motivations affect creepiness and trust in generative artificial intelligence, Telemat. Inf. 83 (2023) 102030.
[2] E. Bagheri, P.G. Esteban, H.L. Cao, A. De Beir, D. Lefeber, B. Vanderborght, An autonomous cognitive empathy model responsive to users' facial emotion expressions, ACM Trans. Interact. Intell. Syst. 10 (3) (2020) 23.
[3] K.S. Barber, J. Kim, Belief Revision Process Based on Trust: Agents Evaluating Reputation of Information Sources. In: Falcone, R., Singh, M., Tan, YH. (eds) Trust in Cyber-societies. Lecture Notes in Computer Science, vol 2246. Springer, Berlin, Heidelberg (2001). https://doi.org/10.1007/3-540-45547-7_5.
[4] P. Bedué, A. Fritzsche, Can we trust AI? An empirical investigation of trust requirements and guide to successful AI adoption, J. Enterp. Inf. Manag. 35 (2) (2022) 530–549.
[5] K.T. Bochniarz, S.K. Czerwiński, A. Sawicki, P.A. Atroszko, Attitudes to AI among high school students: Understanding distrust towards humans will not help us understand distrust towards AI, Pers. Individ. Differ. 185 (2022) 0191–8869, https://doi.org/10.1016/j.paid.2021.111299.
[6] N.G. Bourbakis, G.A. Tsihrintzis, M. Virvou, 14th International Conference on Information, Intelligence, Systems & Applications, IISA 2023, Volos, Greece, July 10-12, 2023. IEEE (2023), ISBN 979-8-3503-1806-7.
[7] K. Chrysafiadi, M. Virvou, G.A. Tsihrintzis, A fuzzy-based mechanism for automatic personalized assessment in an e-learning system for computer programming, Intell. Decis. Technol. 16 (4) (2022) 699–714.
[8] M.C. Dorneich, R. Dudley, E. Letsu-Dake, W. Rogers, S.D. Whitlow, M.C. Dillard, E. Nelson, Interaction of automation visibility and information quality in flight deck information automation, IEEE Trans. Hum.-Mach. Syst. 47 (2017) 915–926, https://doi.org/10.1109/THMS.2017.2717939.
[9] R. Falcone, C. Castelfranchi, Social trust: A cognitive approach. Trust and deception in virtual societies, Springer, 2001.
[10] H. Fang, X. Li, J. Zhang, Integrating social influence modeling and user modeling for trust prediction in signed networks, Artif. Intell. 302 (2022) 103628.
[11] A. Fasoulis, M. Virvou, G.A. Tsihrintzis, C. Patsakis, E. Alepis, Sensus Vox: Sentiment Mapping Through Smartphone Multi-Sensory Crowdsourcing. IEEE 30th International Conference on Tools with Artificial Intelligence, ICTAI 2018, 5-7 November 2018, Volos, Greece. IEEE 2018, ICTAI (2018): 438-444.
[12] D. Gambetta, Trust, Basil Blackwell, Oxford, 1990.
[13] Y. Gao, W. Tong, E.Q. Wu, W. Chen, G. Zhu, F.-Y. Wang, Chat with ChatGPT on interactive engines for intelligent driving, IEEE Trans. Intell. Veh. 8 (3) (2023) 2034–2036.
[14] V. Gherheș, Why are we afraid of artificial intelligence? Eur. Rev. Appl. Sociol. 11 (17) (2018) 6–15, https://doi.org/10.1515/eras-2018-0006.
[15] L. Glikson, W.A. Williams, Human trust in artificial intelligence: review of empirical research, Acad. Manag. Ann. 14 (2) (2020) 627–660, https://doi.org/10.5465/annals.2018.0057.
[16] W. Han, Y. Li, Z. Zhang, C. Xia, STP-based verification and synthesis of state opacity for logical finite state machines, Inf. Sci. 641 (2023) 119130.
[17] P.A. Hancock, D.R. Billings, K.E. Schaefer, J.Y. Chen, E.J. De Visser, R. Parasuraman, A meta-analysis of factors affecting trust in human- robot interaction, Hum. Factors. 53 (5) (2011) 517–527, https://doi.org/10.1177/0018720811417254.
[18] M. Hasnain, M.F. Pasha, I. Ghani, M. Imran, M.Y. Alzahrani, R. Budiarto, Evaluating trust prediction and confusion matrix measures for web services ranking, IEEE Access 8 (2020) 90847–90861, https://doi.org/10.1109/ACCESS.2020.2994222.
[19] R. Howard, S. Barrett, L. Kunze, Don't blindly trust your CNN: Towards competency-aware object detection by evaluating novelty in open-ended environments, in: *2021 IEEE International Conference on Robotics and Automation (ICRA)*, Xi'an, China, 2021, pp. 13286-13292, doi: 10.1109/ICRA48506.2021.9562116.
[20] D. Johnson, K. Graysonb, Cognitive and affective trust in service relationships, J. Bus. Res. 58 (2005) 500–507.
[21] J. Khan, S. Lee, Implicit user trust modeling based on user attributes and behavior in online social networks, IEEE Access 7 (2019) 142826–142842.
[22] A.D. Kaplan, T.T. Kessler, J.C. Brill, P.A. Hancock, Trust in artificial intelligence: meta-analytic findings, Hum. Factors 65 (2) (2023) 337–359.
[23] K. Katsuragawa, A. Kamal, Q.F. Liu, M. Negulescu, E. Lank. Bi-level thresholding: analyzing the effect of repeated errors in gesture input. ACM Trans. Interact. Intell. Syst. 9, 2–3 (2019) 15:1-15:30. https://doi.org/ 10.1145/3181672.
[24] M. Kraus, R. Riekenbrauck, W. Minker, Development of a trust-aware user simulator for statistical proactive dialog modeling in human-AI teams. Adjunct Proceedings of the 31st ACM Conference on User Modeling, Adaptation and Personalization, UMAP 2023, 2023.
[25] M. Kraus, N. Wagner, W. Minker, Effects of Proactive Dialogue Strategies on Human-Computer Trust. Proceedings of the 28th ACM Conference on User Modeling, Adaptation and Personalization, UMAP 2020, Genoa, Italy, July 12-18, 2020. ACM 2020, ISBN 978-1-4503-6861-2 UMAP 2020: 107-116.
[26] B. Lafreniere, T.R. Jonker, S. Santosa, M. Parent, M. Glueck, T. Grossman, H. Benko, D. Wigdor, False positives vs. false negatives: the effects of recovery time and cognitive costs on input error preference, in: The 34th Annual ACM Symposium on User Interface Software and Technology, Virtual Event, 2021, p. pp..
[27] N.K. Lankton, D.H. McKnight, J. Tripp, Technology, humanness, and trust: rethinking trust in technology, J. Assoc. Inf. Syst. 16 (10) (2015) 880–918, https://doi.org/10.17705/1jais.00411.
[28] J.D. Lee, K.A. See, Trust in automation: designing for appropriate reliance, Hum. Factors 46 (1) (2004) 50–80, https://doi.org/10.1518/hfes.46.1.50_30392.
[29] Y. Liang, D. Zou, H. Xie, F.L. Wang, Exploring the potential of using ChatGPT in physics education, Smart Learning Environments 10 (1) (2023) 52.
[30] N. Luhmann, Trust and Power, John Wiley & Sons, Chichester, 2018.
[31] S. Ma, J. Yuan, Z. Guo, Q. Wu, Autonomous plume Near-Source search assisted by intermittent visible plume information using finite state Machine and YOLOv3-tiny, Expert Syst. Appl. 228 (2023) 120350.
[32] C. Martínez-Cruz, C. Porcel, J. Bernabé-Moreno, E. Herrera-Viedma, A model to represent users trust in recommender systems using ontologies and fuzzy linguistic modeling, Inf. Sci. 311 (2015) 102–118.
[33] R.C. Mayer, J.H. Davis, F.D. Schoorman, An integrative model of organizational trust, Acad. Manag. Rev. 20 (3) (1995) 709, https://doi.org/10.2307/258792.
[34] D.H. McKnight, V. Choudhury, C. Kacmar, The impact of initial consumer trust on intentions to transact with a web site: a trust building model, J. Strateg. Inf. Syst. 11 (3–4) (2002) 297–323, https://doi.org/10.1016/S0963-8687(02)00020-3.
[35] B.M. Muir, Trust between humans and machines, and the design of decision aids, Int. J. Man-Machine Stud. 27 (5–6) (1987) 527–539, https://doi.org/10.1016/s0020- 7373(87)80013-5.
[36] Open AI, ChatGPT: OpenAI's Conversational Language Model (2021).
[37] D.P. Panagoulias, F.A. Palamidas, M. Virvou, G.A. Tsihrintzis, Evaluation of ChatGPT-supported diagnosis, staging and treatment planning for the case of lung cancer, in: 2023 20th ACS/IEEE International Conference on Computer Systems and Applications (AICCSA), IEEE, 2023, pp. 1–8.
[38] D.P. Panagoulias, M. Virvou, G.A. Tsihrintzis, A novel framework for artificial intelligence explainability via the Technology Acceptance Model and Rapid Estimate of Adult Literacy in Medicine using machine learning, Expert Syst. Appl. 248 (2024) 123375.
[39] I. Rahwan, M. Cebrian, N. Obradovich, J. Bongard, J.F. Bonnefon, C. Breazeal, J.W. Crandall, N.A. Christakis, I.D. Couzin, M.O. Jackson, N.R. Jennings, E. Kamar, I.M. Kloumann, H. Larochelle, D. Lazer, R. McElreath, A. Mislove, D.C. Parkes, A.S. Pentland, M.E. Roberts, M. Wellman, Machine behaviour, Nature 568 (7753) (2019) 477–486.
[40] T.J. Bodnar, C.S. Tucker, K.M. Hopkinson, S.G. Bilén, Increasing the veracity of event detection on social media networks through user trust modeling, IEEE BigData (2014:) 636–643.

[41] S. Tolmeijer, A. Weiss, M. Hanheide, F. Lindner, T.M. Powers, C. Dixon, M.L. Tielman, Taxonomy of trust-relevant failures and mitigation strategies, in: Proceedings of the 2020 ACM/IEEE International Conference on Human-Robot Interaction, Association for Computing Machinery, 2020, pp. 3–12, https://doi.org/10.1145/3319502.3374793.
[42] G.A. Tsihrintzis, M. Virvou, I. Hatzilygeroudis, Special Collection of Extended Selected Papers on Novel Research Results Presented in The 12th International Conference on Information, Intelligence, Systems and Applications (IISA2021), Chania, Greece, 12-14 July 2021. Intell. Decis. Technol. 15(4) (2021) 641–643.
[43] Y. Ye, H. You, E.J. Du, Improved trust in human-robot collaboration with ChatGPT, IEEE Access 11 (2023) 55748–55754.
[44] D. Ullrich, A. Butz, S. Diefenbach, The development of overtrust: an empirical simulation and psychological analysis in the context of human-robot interaction, Front. Robot. AI 8 (2021) 554578.
[45] J. Vallverdú, Approximate and situated causality in deep learning, Philosophies 5 (2) (2020) 1–12, https://doi.org/10.3390/philosophies5010002.
[46] M. Virvou, G.A. Tsihrintzis, D. Sotiropoulos, K. Chrysafiadi, E. Sakkopoulos, E.-A. Tsichrintzi, ChatGPT in artificial intelligence empowered E-learning for cultural heritage: the case of lyrics and poems. Proceedings of the 14th International Conference on Information, Intelligence, Systems & Applications, IEEE, 2023.
[47] M. Virvou, G.A. Tsihrintzis, Is ChatGPT beneficial to education? A holistic evaluation framework based on intelligent tutoring systems. Proceedings of the 14th International Conference on Information, Intelligence, Systems & Applications, IEEE, 2023.
[48] M. Virvou, Artificial Intelligence and User Experience in reciprocity: Contributions and state of the art, Intell. Decis. Technol. 17 (1) (2023) 73–125.
[49] J. Wang, F. Shahzad, S.F. Ashraf, Elements of information ecosystems stimulating the online consumer behavior: A mediating role of cognitive and affective trust, Telematics Inform. 80 (2023).
[50] J. Zhou, T. Li, S.J. Fong, N. Dey, R.G. Crespo, Exploring ChatGPT's potential for consultation, recommendations and report diagnosis: gastric cancer and gastroscopy reports' case, Int. J. Interact. Multimed. Artif. Intell. 8 (2) (2023) 7.